

Discrete Choice Methods with Simulation

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Please contact me with any corrections, comments,
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to
Daniel McFadden
and
in memory of
Kenneth Train, Sr.

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Chapter 1

Introduction

1.1 Motivation

When I wrote my first book, *Qualitative Choice Analysis*, in the mid 1980's, the field had reached a critical juncture. The break-through concepts that defined the field had been made. The basic models — mainly logit and nested logit — had been introduced, and the statistical and economic properties of these models had been derived. Applications had proven successful in many different areas, including transportation, energy, housing, and marketing — to name only a few.

The field is at a similar juncture today for a new generation of procedures. The first-generation models contained important limitations that inhibited their applicability and realism. These limitations were well recognized at the time, but ways to overcome them had not yet been discovered. Over the past twenty years, tremendous progress has been made, leading to what can only be called a sea-change in the approach and methods of choice analysis. The early models have now been supplemented by a variety of more powerful and more flexible methods. The new concepts have arisen gradually, with researchers building on the work of others. However, in a sense, the change has been more like a quantum leap than a gradual progression. The ways that researchers think about, specify, and estimate their models has changed. Importantly, a kind of consensus, or understanding, seems to have emerged about the new methodology. Among researchers working in the field, a definite sense of purpose and progress prevails.

My purpose in writing this new book is to bring these ideas together, in a form that exemplifies the unity of approach that I feel

has emerged, and in a format that makes the methods accessible to a wide audience. The advances have mostly centered on simulation. Essentially, simulation is the researcher's response to the inability of computers to perform integration. Stated more precisely, simulation provides a numerical approximation to integrals, with different methods offering different properties and being applicable to different kinds of integrands.

Simulation allows estimation of otherwise intractable models. Practically any model can be estimated by some form of simulation. The researcher is therefore freed from previous constraints on model specification – constraints that reflected mathematical convenience rather than the economic reality of the situation. This new flexibility is a tremendous boon to research. It allows more realistic representation of the hugely varied choice situations that arise in the world. It enables the researcher to obtain more information from any given dataset and, in many cases, allows previously unapproachable issues to be addressed.

This flexibility places a new burden on the researcher. First, the methods themselves are more complicated than earlier procedures and utilize many concepts and procedures that are not covered in standard econometrics courses. Understanding the various techniques — their advantages and limitations, and the relations among them — is important when choosing the appropriate method in any particular application and for developing new methods when none of the existing models seems right. The purpose of this book is to assist readers along this path.

Second, to implement a new method, or a variant on an old method, the researcher needs to be able to program the procedure into computer software. This means that the researcher will often need to know how maximum likelihood and other estimation methods work from a computational perspective, how to code specific models, and how to take existing code and change it to represent variations in behavior. Some models, such as mixed logit and pure probit in addition of course to standard logit, are available in commercially available statistical packages. In fact, code for these and other models, as well as manuals and sample data, are available (free) at my website <http://elsa.berkeley.edu/~train>. Whenever appropriate, researchers should use available codes rather than writing their own. However, the true value of the new approach to choice modeling is the ability to

create tailor-made models. The computational and programming steps that are needed to implement a new model are usually not difficult. An important goal of the book is to teach these skills as an integral part of the exposition of the models themselves. I personally find programming to be extremely valuable pedagogically. The process of coding a model helps me to understand how exactly the model operates, the reasons and implications of its structure, what features constitute the essential elements that cannot be changed while maintaining the basic approach, and which features are arbitrary and can easily be changed. I imagine other people learn this way too.

1.2 Choice Probabilities and Integration

To focus ideas, I will now establish the conceptual basis for discrete choice models and show where integration comes into play. An agent (i.e., person, firm, decision-maker) faces a choice, or a series of choices over time, among a set of options. For example, a customer chooses which of several competing products to buy; a firm decides which technology to use in production; a student chooses which answer to give on a multiple choice test; a survey respondent chooses an integer between 1 and 5 on a Lickert-scale question; a worker chooses whether to continue working each year or retire. Denote the outcome of the decision(s) in any given situation as y , where y indicates the chosen option or sequence of options. We assume for the purposes of this book that the outcome variable is discrete in that it takes a countable number of values. Many of the concepts that we describe are easily transferable to situations where the outcome variable is continuous. However, notation and terminology is different with continuous outcome variables than discrete ones. Also, discrete choices generally reveal less information about the choice process than continuous-outcome choices, such that the econometrics of discrete choice is usually more challenging.

Our goal is to understand the behavioral process that leads to the agent's choice. We take a causal perspective. There are factors that collectively determine, or cause, the agent's choice. Some of these factors are observed by the researcher and some are not. The observed factors are labeled x and the unobserved factors ε . The factors relate to the agent's choice through a function $y = h(x, \varepsilon)$. This function is called the behavioral process. It is deterministic in the sense that given x and ε , the choice of the agent is fully determined.

Since ε is not observed, the agent's choice is not deterministic and cannot be predicted exactly. Instead, the *probability* of any particular outcome is derived. The unobserved terms are considered random with density $f(\varepsilon)$. The probability that the agent chooses a particular outcome from the set of all possible outcomes is simply the probability that the unobserved factors are such that the behavioral process results in that outcome: $P(y|x) = \text{Prob}(\varepsilon \text{ s.t. } h(x, \varepsilon) = y)$.

We can express this probability in a more useable form. Define an indicator function $I[h(x, \varepsilon) = y]$ that takes the value of 1 when the statement in brackets is true and 0 when the statement is false. That is, $I[\cdot] = 1$ if the value of ε , combined with x , induces the agent to choose outcome y and 0 if the value of ε , combined with x , induces the agent to choose some other outcome. Then the probability that the agent chooses outcome y is simply the expected value of this indicator function, where the expectation is over all possible values of the unobserved factors:

$$\begin{aligned} P(y|x) &= \text{Prob}(I[h(x, \varepsilon) = y] = 1) \\ &= \int I[h(x, \varepsilon) = y] f(\varepsilon) d\varepsilon. \end{aligned} \quad (1.1)$$

Stated in this form, the probability is an integral — specifically an integral of an indicator for the outcome of the behavioral process over all possible values of the unobserved factors.

To calculate this probability, the integral must be evaluated. There are three possibilities.

Complete closed-form expression

For certain specifications of h and f , the integral can be expressed in closed form. In these cases, the choice probability can be calculated exactly from the closed-form formula. For example, consider a binary logit model of whether or not a person takes a given action, such as buying a new product. The behavioral model is specified as follows. The person would obtain some net benefit or utility from taking the action. This utility, which can be either positive or negative, consists of a part that is observed by the researcher, $\beta'x$ where x is a vector of variables and β is a vector of parameters, and a part that is not observed, ε : $U = \beta'x + \varepsilon$. The person takes the action only if the utility is positive, that is, only if doing so provides a net benefit. The

probability that the person takes the action, given what the researcher can observe, is therefore $P = \int I[\beta'x + \varepsilon > 0]f(\varepsilon)d\varepsilon$ where f is the density of ε . Assume that ε is distributed logistic, such that its density is $f(\varepsilon) = e^{-\varepsilon}/(1 + e^{-\varepsilon})^2$ with cumulative distribution $F(\varepsilon) = 1/(1 + e^{-\varepsilon})$. Then the probability of the person taking the action is:

$$\begin{aligned}
 P &= \int I[\beta'x + \varepsilon > 0]f(\varepsilon)d\varepsilon \\
 &= \int I[\varepsilon > -\beta'x]f(\varepsilon)d\varepsilon \\
 &= \int_{\varepsilon=-\beta'x}^{\infty} f(\varepsilon)d\varepsilon \\
 &= 1 - F(-\beta'x) = 1 - \frac{1}{1 + e^{\beta'x}} \\
 &= \frac{e^{\beta'x}}{1 + e^{\beta'x}}
 \end{aligned}$$

For any x , the probability can be calculated exactly as $P = \exp(\beta'x)/(1 + \exp(\beta'x))$.

Other models also have closed-form expressions for the probabilities. Multinomial logit (in chapter 3), nested logit (chapter 4), and ordered logit (chapter 7) are prominent examples. The methods that I described in my first book and that served as the basis for the first wave of interest in discrete choice analysis relied almost exclusively on models with closed-form expressions for the choice probabilities. In general, however, the integral for probabilities cannot be expressed in closed form. More to the point, restrictions must be placed on the behavioral model h and the distribution of random terms f in order for the integral to take a closed form. These restrictions can make the models unrealistic for many situations.

Complete simulation

Rather than solve the integral analytically, it can be approximated through simulation. Simulation is applicable in one form or another to practically any specification of h and f . Simulation relies on the fact that integration over a density is a form of averaging. Consider the integral $\bar{t} = \int t(\varepsilon)f(\varepsilon)d\varepsilon$ where $t(\varepsilon)$ is a statistic based on ε which has density $f(\varepsilon)$. This integral is the expected value of t over all possible values of ε . This average can be approximated in an intuitively

straightforward way. Take numerous draws of ε from its distribution f , calculate $t(\varepsilon)$ for each draw, and average the results. This simulated average is an unbiased estimate of the true average. It approaches the true average as more and more draws are used in the simulation.

This concept of simulating an average is the basis for all simulation methods, at least all of those that we consider in this book. As given in eq. (1.1), the probability of a particular outcome is an average of the indicator $I(\cdot)$ over all possible values of ε . The probability, when expressed in this form, can be simulated directly as follows: (1) Take a draw of ε from $f(\varepsilon)$. Label this draw ε^1 , where the superscript denotes that it is the first draw. (2) Determine whether $h(x, \varepsilon^1) = y$ with this value of ε . If so, create $I^1 = 1$, otherwise set $I^1 = 0$. (3) Repeat steps 1 and 2 many times, for a total of R draws. The indicator for each draw is labeled I^r for $r = 1, \dots, R$. (4) Calculate the average of the I^r 's. This average is the simulated probability: $\check{P}(i | x) = \frac{1}{R} \sum_{r=1}^R I^r$. It is the proportion of times that the draws of the unobserved factors, when combined with the observed variables x , result in outcome y .

As we will see in the chapters to follow, this simulator, while easy to understand, has some unfortunate properties. Choice probabilities can often be expressed as averages of other statistics, rather than the average of an indicator function. The simulators based on these other statistics are calculated analogously, by taking draws from the density, calculating the statistic and averaging the results. Probit (in chapter 5) is the most prominent example of a model estimated by complete simulation. Various methods of simulating the probit probabilities have been developed based on averages of various statistics over various (related) densities.

Partial simulation/partial closed form

So far we have provided two polar extremes: either solve the integral analytically or simulate it. In many situations, it is possible to do some of both.

Suppose the random terms can be decomposed into two parts labeled ε_1 and ε_2 . Let the joint density of ε_1 and ε_2 be $f(\varepsilon) = f(\varepsilon_1, \varepsilon_2)$. The joint density can be expressed as the product of a marginal and a conditional density: $f(\varepsilon_1, \varepsilon_2) = f(\varepsilon_2 | \varepsilon_1) \cdot f(\varepsilon_1)$. With this decomposition, the probability in eq. (1.1) can be expressed as:

$$P(y|x) = \int I[h(x, \varepsilon) = y] f(\varepsilon) d\varepsilon$$

$$= \int_{\varepsilon_1} \left[\int_{\varepsilon_2} I[h(x, \varepsilon_1, \varepsilon_2) = y] f(\varepsilon_2 | \varepsilon_1) d\varepsilon_2 \right] f(\varepsilon_1) d\varepsilon_1.$$

Now suppose that a closed form exists for the integral in large brackets. Label this formula $g(\varepsilon_1) \equiv \int_{\varepsilon_2} I[h(x, \varepsilon_1, \varepsilon_2) = y] f(\varepsilon_2 | \varepsilon_1) d\varepsilon_2$, which is conditional on the value of ε_1 . The probability then becomes $P(y|x) = \int_{\varepsilon_1} g(\varepsilon_1) f(\varepsilon_1) d\varepsilon_1$. If a closed-form solution does not exist for this integral, then it is approximated through simulation. Note that it is simply the average of g over the marginal density of ε_1 . The probability is simulated by taking draws from $f(\varepsilon_1)$, calculating $g(\varepsilon_1)$ for each draw, and averaging the results.

This procedure is called “convenient error partitioning” (Train, 1998). The integral over ε_2 given ε_1 is calculated exactly while the integral over ε_1 is simulated. There are clear advantages to this approach over complete simulation. Analytic integrals are both more accurate and easier to calculate than simulated integrals. It is useful, therefore, when possible, to decompose the random terms such that some of them can be integrated analytically, even if the rest must be simulated. Mixed logit (in chapter 6) is a prominent example of a model that uses this decomposition effectively. Other examples include Gourieroux and Monfort’s (1993) binary probit model on panel data and Bhat’s (1999) analysis of ordered responses.

1.3 Outline of book

Discrete choices analysis consists of two interrelated tasks: specification of the behavioral model and estimation of the parameters of that model. Simulation plays a part in both tasks. Simulation allows the researcher to approximate the choice probabilities that arise in the behavioral model. As we have stated, the ability to use simulation frees the researcher to specify models without the constraint that the resulting probabilities have a closed form. Simulation also enters the estimation task. The properties of an estimator, such as maximum likelihood, can change when simulated probabilities are used instead of the actual probabilities. Understanding these changes, and mitigating any ill effects, is important for a researcher. In some cases, such as with Bayesian procedures, the estimator itself is an integral over a density (as opposed to the choice probability being an integral). Simulation allows these estimators to be implemented even when the integral that defines the estimator does not take a closed form.

The book is organized around these two tasks. Part I describes behavioral models that have been proposed to describe the choice process. The chapters in this section move from the simplest model, logit, to progressively more general and consequently more complex models. A chapter is devoted to each of: logit, the family of generalized extreme value models (whose most prominent member is nested logit), probit, and mixed logit. This part of the book ends with a chapter entitled “variations on a theme,” which covers a variety of models that build upon the concepts in the previous chapters. The point of this chapter is more than simply to introduce various new models. The chapter illustrates the underlying concept of the book, namely, that researchers need not rely on the few common specifications that have been programmed into software but can design models that reflect the unique setting, data and goals of their project, writing their own software and using simulation as needed.

Part II describes estimation of the behavioral models. Numerical maximization is covered first, since most estimation procedures involve maximization of some function, such as the log-likelihood function. We then describe procedures for taking draws from various kinds of densities, which is the basis for simulation. This chapter also describes different kinds of draws, including antithetic variants and quasi-random sequences, that can provide greater simulation accuracy than independent random draws. We then turn to simulation-assisted estimation, looking first at classical procedures, including maximum simulated likelihood, method of simulated moments, and method of simulated scores. Finally, we examine Bayesian estimation procedures, which use simulation to approximate moments of the posterior distribution. The Bayesian estimator can be interpreted from either a Bayesian or classical perspective and has the advantage of avoiding some of the numerical difficulties associated with classical estimators. The power that simulation provides when coupled with Bayesian procedures makes this chapter a fitting finale for the book.

1.4 Topics not covered

I feel it is useful to say a few words about what the book does not cover. There are several topics that could logically be included but are not. One is the branch of empirical industrial organization that involves estimation of discrete choice models of consumer demand on market-

level data. Customer-level demand is specified by a discrete choice model, such as logit or mixed logit. This formula for customer-level demand is aggregated over consumers to obtain market-level demand functions that relate prices to shares. Market equilibrium prices are determined as the interaction of these demand functions with supply, based on marginal costs and the game that the firms are assumed to play. Berry (1994) and Berry, Levinsohn and Pakes (1995) developed methods for estimating the demand parameters when the customer-level model takes a flexible form such as mixed logit. The procedure has been implemented in numerous markets for differentiated goods, such as ready-to-eat cereals (Nevo, 2001).

I have decided not to cover these procedures, despite their importance, because they are very different, in motivation, data, and methodology, than all the other material in the book. They use market-level data rather than customer-level. The observed demand is a set of continuous variables, representing market shares, rather than discrete variables representing individual customers' choices. Prices are endogenous, determined by the interaction of demand and supply. In contrast, prices faced by an individual customer are not affected by that customer's choice; prices are therefore exogenous from market forces when modeling customer-level demand. Other types of endogeneity may arise with customer-level data, but the endogeneity from demand/supply interaction, which market level data embody, does not.

The focus of the market-level methods, and the insight provided by the contributors to these methods, are different from those for customer-level data. Instruments become paramount since price is endogenous. The dominant questions, and the basis for much of the discussion in the field, concern the selection of instruments: what instruments are appropriate and what identifying assumptions are necessary with any given instruments? These questions do not arise with customer-level data (unless, of course, some other form of endogeneity is present). The dominant methodological issue is also different. Since instruments are necessary, the methodological question becomes: how can market share data, which arise from demand functions in which the unobserved factors enter nonlinearly, be transformed to allow the use of instrumental variables estimation, which operates effectively when the unobserved factors enter linearly? Berry *et al.*'s contribution was in determining the appropriate transformation and how to perform it. This issue does not arise in customer-level data. For interested read-

ers, Nevo (2000) provides a useful guide to the procedures, including programming instructions and an discussion of the appropriate choice of instruments.

A second area that this book does not cover is discrete/continuous models. These models arise when a regression equation for a continuous variable is related in any of several ways to a discrete choice. The most prominent situations are the following.

1. The continuous variable depends on a discrete explanatory variable that is determined endogenously with the dependent variable. For example, consider an analysis of the impact of job-training programs on wages. A regression equation is specified with wages as the dependent variable and a dummy variable for whether the person participated in a job-training program. The coefficient of the participation dummy indicates the impact of the program on wages. The situation is complicated, however, by the fact that participation is voluntary: people choose whether to participate in job-training programs. The decision to participate is at least partially determined by factors that also affect the person's wage, such as the innate drive, or "go-for-it" attitude, of the person. Estimation of the regression by ordinary least squares is biased in this situation, since the program-participation dummy is correlated with the errors in the wage equation.
2. A regression equation is estimated on a sample of observations that are selected on the basis of a discrete choice that is determined endogenously with the dependent variable. For example, a researcher might want to estimate the impact of weather on peak energy load (that is, consumption during the highest-demand hour of the day). Data on energy loads by time of day are only available only for households that have chosen time-of-use rates. However, the households' choice of rate plan can be expected to be related to their energy consumption, with customers who have high peak loads tending not to choose time-of-use rates, since those rates charge high prices in the peak. Estimation of the regression equation on this "self-selected" sample is biased unless the endogeneity of the sample is accounted for.
3. The continuous dependent variable is truncated. For example, consumption of goods by households is necessarily positive. Stated

statistically, consumption is truncated below at zero, and for many goods (such as opera tickets) observed consumption is at this truncation point for a large share of the population. Estimation of the regression without regard to the truncation can cause bias.

The initial concepts regarding appropriate treatment of continuous/discrete models were developed by Heckman (1978, 1979) and Dubin and McFadden (1984). These early concepts are covered in my earlier book (Train, 1986, Ch. 5). Since then, the field has expanded tremendously. An adequate discussion of the issues and procedures would take a book in itself. Moreover, the field has not reached (at least in my view) the same type of juncture that discrete choice modeling has reached. Many fundamental concepts are still being hotly debated, and potentially valuable new procedures have been introduced so recently that there has not been an opportunity for researchers to test them in a variety of settings. The field is still expanding more than it is coalescing.

There are several on-going directions of research in this area. The early procedures were highly dependent on distributional assumptions that are hard to verify. Researchers have been developing semi- and non-parametric procedures that are hopefully more robust. The special 1986 issue of the *Journal of Econometrics* provides a set of important articles on the topic. Papers by Lewbel and Linton (forthcoming) and Levy (2001) describe more recent developments. Another important development concerns the representation of behavior in these settings. The relation between the discrete and continuous variables has been generalized beyond the fairly simple representation that the early methods assumed. For example, in the context of job training, it is likely that the impact of the training differs over people and that people choose to participate in the training program on the basis of the impact it will have on them. Stated in econometric terms: the coefficient of the participation dummy in the wage equation varies over people and affects the value of the dummy. The dummy is correlated with its own coefficient, as well as with the unobserved variables that enter the error of the regression. A recent discussion of approaches to this issue is provided by Carneiro *et al.* (2001).

1.5 A couple notes

Throughout the book, I refer to the researcher as "she" and the decision-maker as "he." This usage, as well as being comparatively gender neutral (or at least symmetrically non-inclusive), allows both people to be referred to in the same paragraph without confusion.

Many colleagues have provided valuable comments and suggestions on earlier drafts of the book. I am very grateful for this help. I thank: Greg Allenby, Moshe Ben-Akiva, Chandra Bhat, Denis Bolduc, David Brownstone, Siddhartha Chib, Jon Eisen-Hecht, Florian Heiss, David Hensher, Joe Herriges, Rich Johnson, Frank Koppelman, Jordan Louviere, Aviv Nevo, Juan de Dios Ortuzar, Ken Small, Joan Walker, Cliff Winston, Joachim Winter, and the students in my graduate econometrics course.

I welcome readers to contact me if you feel I have not covered material that you consider important, or if I have confused rather than enlightened any of the material that I *do* cover. Hopefully, another edition of this book will someday materialize.

Part I

Behavioral Models

Chapter 2

Properties of Discrete Choice Models

2.1 Overview

This chapter describes the features that are common to all discrete choice models. We start by discussing the choice set, which is the set of options that are available to the decision-maker. We then define choice probabilities and derive them from utility maximizing behavior. The most prominent types of discrete choice models, namely logit, GEV, probit, and mixed logit, are introduced and compared within the context of this general derivation. Utility, as a constructed measure of well-being, has no natural level or scale. This fact has important implications for the specification and normalization of discrete choice models, which we explore. We then show how individual-level models are aggregated to obtain market-level predictions, and how the models are used for forecasting over time.

2.2 The choice set

Discrete choice models describe decision-makers' choices among alternatives. The decision-makers can be people, households, firms, or any other decision-making unit, and the alternatives might represent competing products, courses of action, or any other options or items over which choices must be made. To fit within a discrete choice framework, the set of alternatives, called "the choice set," needs to exhibit three

characteristics. First, the alternatives must be *mutually exclusive* from the decision-maker's perspective. Choosing one alternative necessarily implies not choosing any of the other alternatives. The decision-maker chooses only one alternative from the choice set. Second, the choice set must be *exhaustive*, in that all possible alternatives are included. The decision-maker necessarily chooses one of the alternatives. Third, the number of alternatives must be finite. The researcher can count the alternatives and eventually be finished counting.

The first and second criteria are not restrictive. Appropriate definition of alternatives can nearly always assure that the alternatives are mutually exclusive and the choice set is exhaustive. For example, suppose two alternatives labeled A and B are not mutually exclusive because the decision-maker can choose both of the alternatives. The alternatives can be redefined to be "A only," "B only," and "both A and B," which are necessarily mutually exclusive. Similarly, a set of alternatives might not be exhaustive because the decision-maker has the option of not choosing any of them. In this case, an extra alternative can be defined as "none of the other alternatives." The expanded choice set, consisting of the original alternatives plus this new one, is clearly exhaustive.

Often the researcher can satisfy these two conditions in several different ways. The appropriate specification of the choice set in these situations is governed largely by the goals of the research and the data that are available to the researcher. Consider households' choice among heating fuels, a topic which has been studied extensively in efforts to forecast energy use and to develop effective fuel-switching and energy conservation programs. The available fuels are usually natural gas, electricity, oil and wood. These four alternatives, as listed, violate both mutual exclusivity and exhaustiveness. The alternatives are not mutually exclusive because a household can (and many do) have two types of heating, e.g., a natural gas central heater and electric room heaters, or a wood stove along with electric baseboard heating. And the set is not exhaustive because the household can have no heating (which, unfortunately, is not as rare as one might hope.) The researcher can handle each of these issues in several ways. To obtain mutually exclusive alternatives, one approach is to list every possible combination of heating fuels as an alternative. The alternatives are then defined as: "electricity alone," "electricity and natural gas, but no other fuels," and so on. Another approach is to define the choice as the choice among

fuels for the “primary” heating source. Under this procedure, the researcher develops a rule for determining which heating fuel is primary when a household uses multiple heating fuels. By definition, only one fuel (electricity, natural gas, oil, or wood) is primary. The advantage of listing every possible combination of fuels is that it avoids the need to define a “primary” fuel, which is a difficult and somewhat arbitrary distinction. Also, with all combinations considered, the researcher has the ability to examine the factors that determine households’ use of multiple fuels. However, to implement this approach, the researcher needs data that distinguish the alternatives, for example, the cost of heating a house with natural gas and electricity versus the cost with natural gas alone. If the researcher restricts the analysis to choice of primary fuel, then the data requirements are less severe. Only the costs associated with each fuel are needed. Also, a model with four alternatives is inherently easier to estimate and forecast with than a model with the large number of alternatives that arises when every possible combination of fuels is considered. The researcher will need to take these tradeoffs into consideration when specifying the choice set.

The same type of issue arises with regard to exhaustiveness. In our case of heating fuel choice, the researcher can either include “no heating” as an alternative or can redefine the choice situation as being the choice of heating fuel conditional on having heating. The first approach allows the researcher to examine the factors that relate to whether a household has heating. However, this ability is only realized if the researcher has data that meaningfully relate to whether or not a household has heating. Under the second approach, the researcher excludes from the analysis households without heating, and, by doing so, is relieved of the need for data that relate to these households.

As we have just described, the conditions of mutual exclusivity and exhaustiveness can usually be satisfied, and the researcher often has several approaches for doing so. In contrast, the third condition, namely, that the number of alternatives is finite, is actually restrictive. This condition is the defining characteristic of discrete choice models and distinguishes their realm of application from that for regression models. With regression models, the dependent variable is continuous, which means that there is an infinite number of possible outcomes. The outcome might be chosen by a decision-maker, such as the decision of how much money to hold in savings accounts. However, the

alternatives available to the decision-maker, which are every possible monetary value above zero, is not finite (at least not if all fractions are considered, which is an issue we return to below.) When there is an infinite number of alternatives, discrete choice models cannot be applied.

Often regression models and discrete choice models are distinguished by saying that regressions examine choices of “how much” and discrete choice models examine choice of “which.” This distinction, while perhaps illustrative, is not actually accurate. Discrete choice models can and have been used to examine choices of “how much.” A prominent example is households’ choice of how many cars to own. The alternatives are 0, 1, 2, and so on, up to the highest number that the researcher defines as feasible (or observed.) This choice set contains a finite number of mutually exclusive and exhaustive alternatives, such that it is appropriate for analysis via discrete choice models. The researcher can also define the choice set more succinctly as 0, 1, and 2 or more vehicles, if the goals of the research can be met with this specification.

When considered in this way, most choices involving “how many” can be represented in a discrete choice framework. In the case of savings accounts, every one-dollar increment (or even every one-cent increment) can be considered an alternative, and as long as some finite maximum exists, then the choice set fits the criteria for discrete choice. Whether to use regression or discrete choice models in these situations is a specification issue that the researcher must consider. Usually a regression model is more natural and is certainly easier. A discrete choice model would be used in these situations only if there are compelling reasons for doing so. As an example, Train, McFadden and Ben-Akiva (1987) analyzed the number and duration of phone calls that households make, using a discrete choice model instead of a regression model because the discrete choice model allowed greater flexibility in handling the non-linear price schedules that households face. In general, the researcher needs to consider the goals of the research and the capabilities of alternative methods when deciding whether to apply a discrete choice model.

2.3 Derivation of choice probabilities

Discrete choice models are usually derived under an assumption of utility-maximizing behavior by the decision-maker. Thurstone (1927) originally developed the concepts in terms of psychological stimuli, leading to a binary probit model of whether respondents can differentiate the level of stimulus. Marschak (1960) interpreted the stimuli as utility and provided a derivation from utility maximization. Following Marschak, models that can be derived in this way are called random utility models (RUMs). It is important to note, however, that models derived from utility maximization can also be used to represent decision-making that does not entail utility maximization. The derivation assures that the model is consistent with utility maximization; it does not preclude the model from being consistent with other forms of behavior. The models can also be seen as simply describing the relation of explanatory variables to the outcome of a choice, without reference to exactly how the choice is made.

Random utility models are derived as follows. A decision-maker, labeled n , faces a choice among J alternatives. The decision-maker would obtain a certain level of utility (or profit) from each alternative. The utility that decision-maker n obtains from alternative j is U_{nj} , $j = 1, \dots, J$. This utility is known to the decision-maker but not, as we see below, by the researcher. The decision-maker chooses the alternative that provides the greatest utility. The behavioral model is therefore: choose alternative i if and only if $U_{ni} > U_{nj} \quad \forall j \neq i$.

Consider now the researcher. The researcher does not observe the decision-maker's utility. The researcher observes some attributes of the alternatives as faced by the decision-maker, labeled $x_{nj} \quad \forall j$, and some attributes of the decision-maker, labeled s_n , and can specify a function that relates these observed factors to the decision-maker's utility. The function is denoted $V_{nj} = V(x_{nj}, s_n) \quad \forall j$ and is often called "representative utility." Usually, V depends on parameters that are unknown to the researcher and therefore estimated statistically; however, this dependence is suppressed for the moment.

Since there are aspects of utility that the researcher does not or cannot observe, $V_{nj} \neq U_{nj}$. Utility is decomposed as $U_{nj} = V_{nj} + \varepsilon_{nj}$, where ε_{nj} captures the factors that affect utility but are not included in V_{nj} . This decomposition is fully general, since ε_{nj} is defined as simply the difference between true utility U_{nj} and the part of utility that the

researcher captures in V_{nj} . Given its definition, the characteristics of ε_{nj} , such as its distribution, depend critically on the researcher's specification of V_{nj} . In particular, ε_{nj} is not defined for a choice situation *per se*. Rather, it is defined relative to a researcher's representation of that choice situation. This distinction becomes relevant when evaluating the appropriateness of various specific discrete choice models.

The researcher does not know $\varepsilon_{nj} \forall j$ and therefore treats these terms as random. The joint density of the random vector $\varepsilon_n = \langle \varepsilon_{n1}, \dots, \varepsilon_{nJ} \rangle$ is denoted $f(\varepsilon_n)$. With this density, the researcher can make probabilistic statements about the decision-maker's choice. The probability that decision-maker n chooses alternative i is

$$\begin{aligned} P_{ni} &= \text{Prob}(U_{ni} > U_{nj} \forall j \neq i) \\ &= \text{Prob}(V_{ni} + \varepsilon_{ni} > V_{nj} + \varepsilon_{nj} \forall j \neq i) \\ &= \text{Prob}(\varepsilon_{nj} - \varepsilon_{ni} < V_{ni} - V_{nj} \forall j \neq i) \end{aligned} \quad (2.1)$$

This probability is a cumulative distribution, namely, the probability that each random term $\varepsilon_{nj} - \varepsilon_{ni}$ is below the observed quantity $V_{ni} - V_{nj}$. Using the density $f(\varepsilon_n)$, this cumulative probability can be rewritten as:

$$\begin{aligned} P_{ni} &= \text{Prob}(\varepsilon_{nj} - \varepsilon_{ni} < V_{ni} - V_{nj} \forall j \neq i) \\ &= \int_{\varepsilon} I(\varepsilon_{nj} - \varepsilon_{ni} < V_{ni} - V_{nj} \forall j \neq i) f(\varepsilon_n) d\varepsilon_n, \end{aligned} \quad (2.2)$$

where $I(\cdot)$ is the indicator function, equaling 1 when the term in parentheses is true and 0 otherwise. This is a multidimensional integral over the density of the unobserved portion of utility, $f(\varepsilon_n)$. Different discrete choice models are obtained from different specifications of this density, that is, from different assumptions about the distribution of the unobserved portion of utility. The integral takes a closed-form only for certain specifications of $f(\cdot)$. Logit and nested logit have closed-form expressions for this integral. They are derived under the assumption that the unobserved portion of utility is distributed iid extreme value and a type of generalized extreme value, respectively. Probit is derived under the assumption that $f(\cdot)$ is a multivariate normal, and mixed logit is based on the assumption that the unobserved portion of utility consists of a part that follows any distribution specified by the researcher plus a part that is iid extreme value. With probit and mixed logit, the resulting integral does not have a closed form and

is evaluated numerically through simulation. Each of these models is discussed in detail in subsequent chapters.

The meaning of choice probabilities is more subtle, and more revealing, than it might at first appear. An example serves as illustration. Consider a person who can take either a car or bus to work. The researcher observes the time and cost that the person would incur under each mode. However, the researcher realizes that there are factors other than time and cost that affect the person's utility and hence his choice. The researcher specifies

$$\begin{aligned} V_c &= \alpha T_c + \beta M_c \\ V_b &= \alpha T_b + \beta M_b \end{aligned}$$

where T_c and M_c are the time and cost (i.e., money) that the person incurs traveling to work by car, T_b and M_b are defined analogously for bus, and the subscript n denoting the person is omitted for convenience. The coefficients α and β are either known or estimated by the researcher.

Suppose that, given α and β and the researcher's measures of the time and cost by car and bus, it turns out that $V_c = 4$ and $V_b = 3$. This means that, on observed factors, car is better for this person than bus by 1 unit. (We discuss below the normalization of utility that sets the dimension to these units.) It does not mean, however, that the person necessarily chooses car, since there are other factors not observed by the researcher that affect the person. The probability that the person chooses bus instead of car is the probability that the unobserved factors for bus are sufficiently better than those for car to overcome the advantage that car has on observed factors. Specifically, the person will choose bus if the unobserved portion of utility is higher than that for car by at least 1 unit, thus overcoming the 1 unit advantage that car has on observed factors. The probability of this person choosing bus is therefore the probability that $\varepsilon_b - \varepsilon_c > 1$. Conversely, the person will choose car if the unobserved utility for bus is *not* better than that for car by at least 1 unit, that is, if $\varepsilon_b - \varepsilon_c < 1$. Since 1 is the difference between V_c and V_b in our example, the probabilities can be stated more explicitly as:

$$\begin{aligned} P_c &= \text{Prob}(\varepsilon_b - \varepsilon_c < V_c - V_b) \\ \text{and} \\ P_b &= \text{Prob}(\varepsilon_b - \varepsilon_c > V_c - V_b) \end{aligned}$$

$$= \text{Prob}(\varepsilon_c - \varepsilon_b < V_b - V_c).$$

These equations are the same as equation (2.1) above, re-expressed for our car-bus example.

The question arises in the derivation of the choice probabilities: what is meant by the distribution of ε_n ? The interpretation that the researcher places on this density affects the researcher's interpretation of the choice probabilities. The most prominent way to think about this distribution is as follows. Consider a population of people who face the same observed utility $V_{nj} \forall j$ as person n . Among these people, the values of the unobserved factors differ. The density $f(\varepsilon_n)$ is the distribution of the unobserved portion of utility within the population of people who face the same observed portion of utility. Under this interpretation, the probability P_{ni} is the share of people who choose alternative i within the population of people who face the same observed utility for each alternative as person n . The distribution can also be considered in subjective terms, as representing the researcher's subjective probability that the person's unobserved utility will take given values. In this case, the P_{ni} is the probability that the researcher ascribes to the person's choosing alternative i given the researcher's ideas about the unobserved portions of the person's utility. As a third possibility, the distribution can represent the effect of factors that are quixotic to the decision-maker himself (representing, e.g., aspects of bounded rationality), such that P_{ni} is the probability that these quixotic factors induce the person to choose alternative i given the observed, non-quixotic factors.

2.4 Specific models

Logit, GEV, probit, and mixed logit are discussed at length in the subsequent chapters. However, a quick preview of these models is useful at this point, to show how they relate to the general derivation of all choice models and how they differ within this derivation. As stated above, different choice models are derived under different specifications of the density of unobserved factors, $f(\varepsilon_n)$. The issues therefore are: what distribution is assumed for each model, and what is the motivation for these different assumptions.

Logit (discussed in Chapter 3) is by far the most widely used discrete choice model. It is derived under the assumption that ε_{ni} is

distributed iid extreme value for all i . The critical part of the assumption is that the unobserved factors are uncorrelated over alternatives, as well as having the same variance for all alternatives. This assumption, while restrictive, provides a very convenient form for the choice probability. The popularity of the logit model is due to this convenience. However, the assumption of independence can be inappropriate in some situations. Unobserved factors related to one alternative might be similar to those related to another alternative. For example, a person who dislikes travel by bus because of the presence of other riders might have a similar reaction to rail travel; if so, then the unobserved factors affecting bus and rail are correlated rather than independent. The assumption of independence also enters When a logit model is applied to sequences of choices over time. The logit model assumes that each choice is independent of the others. In many cases, one would expect that unobserved factors that affect the choice in one period would persist, at least somewhat, into the next period, inducing dependence among the choices over time.

The development of other models has arisen largely to avoid the independence assumption within a logit. Generalized extreme value models (GEV, discussed in Chapter 4) are based, as the name implies, on a generalization of the extreme value distribution. The generalization can take many forms, but the common element is that it allows correlation in unobserved factors over alternatives and collapses to the logit model when this correlation is zero. Depending on the type of GEV model, the correlations can be more or less flexible. For example, a comparatively simple GEV model places the alternatives into several groups called nests, with unobserved factors having the same correlation for all alternatives within a nest and no correlation for alternatives in different nests. More complex forms allow essentially any pattern of correlation. GEV models usually have closed-forms for the choice probabilities, such that simulation is not required for their estimation.

Probits (Chapter 5) are based on the assumption that the unobserved factors are distributed jointly normal: $\varepsilon'_n = \langle \varepsilon_{n1}, \ldots, \varepsilon_{nJ} \rangle \sim N(0, \Omega)$. With full covariance matrix Ω , any pattern of correlation and heteroskedasticity can be accommodated. When applied to sequences of choices over time, the unobserved factors are assumed to be jointly normal over time as well as over alternatives, with any temporal correlation pattern. The flexibility of the probit model in handling correlations over alternatives and time is its main advantage. Its only

functional limitation is arises from its reliance on the normal distribution. In some situations, unobserved factors might not be normally distributed. For example, customer's willingness to pay for a desirable attribute of a product is necessary positive. Assuming that this unobserved factor is normally distributed contradicts the fact that it is positive, since the normal distribution has density on both sides of zero.

Mixed logit (chapter 6) allows the unobserved factors to follow any distribution. The defining characteristic of a mixed logit is that the unobserved factors can be decomposed into a part that contains all the correlation and heteroskedasticity, and another part that is iid extreme value. The first part can follow any distribution, including non-normal distributions. We will show that mixed logit can approximate any discrete choice model and, as such, is fully general.

Other discrete choice models (Chapter 7) have been specified by researchers for specific purposes. Often these models are obtained by combining concepts from other models. For example, a mixed probit is obtained by decomposing the unobserved factors into two parts, as in mixed logit, but giving the second part a normal distribution instead of extreme value. This model has the generality of mixed logit and yet for some situations can be easier to estimate. By understanding the derivation and motivation for all the models, each researcher can specify a model that is tailor-made for the situation and goals of her research.

2.5 Identification of choice models

Several aspects of the behavioral decision process affect the specification and estimation of any discrete choice model. The issues can be summarized easily in two statements: "Only differences in utility matter" and "The scale of utility is arbitrary." The implications of these statements are far-reaching, subtle and, in many cases, quite complex. We discuss them below.

2.5.1 Only differences in utility matter

The absolute level of utility is irrelevant to both the decision-maker's behavior and the researcher's model. If a constant is added to the utility of all alternatives, the alternative with the highest utility doesn't

change. The decision-maker chooses the same alternative with $U_{nj} \forall j$ as with $U_{nj} + k \forall j$ for any constant k . A colloquial way to express this fact is, “A rising tide raises all boats.”

The level of utility doesn’t matter from the researcher’s perspective either. The choice probability is $P_{ni} = \text{Prob}(U_{ni} > U_{nj} \forall j \neq i) = \text{Prob}(U_{ni} - U_{nj} > 0 \forall j \neq i)$, which depends only on the difference in utility, not its absolute level. When utility is decomposed into the observed and unobserved parts, equation (2.1) expresses the choice probability as $P_{ni} = \text{Prob}(\varepsilon_{nj} - \varepsilon_{ni} < V_{ni} - V_{nj} \forall j \neq i)$, which also depends only on differences.

The fact that only differences in utility matter has several implications for the identification and specification of discrete choice models. In general it means that the only parameters that can be estimated (that is, are identified) are those that capture differences across alternatives. This general statement takes several forms.

Alternative-specific constants

It is often reasonable to specify the observed part of utility to be linear in parameters with a constant: $V_{nj} = x'_{nj}\beta + k_j, \forall j$, where x_{nj} is a vector of variables that relate to alternative j as faced by decision-maker n , β are coefficients of these variables, and k_j is a constant that is specific to alternative j . The alternative-specific constant for an alternative captures the average impact on utility of all factors that are not included in the model. As such, they serve a similar function as the constant in a regression model, which also captures the average impact of all unincluded factors.

When alternative-specific constants are included, the unobserved portion of utility, ε_{nj} , has zero mean by construction. If ε_{nj} has a nonzero mean when the constants are not included, then adding the constants makes the remaining error have zero mean: that is, if $U_{nj} = x'_{nj}\beta + \varepsilon_{nj}^*$ with $E(\varepsilon_{nj}^*) = k_j \neq 0$, then $U_{nj} = x'_{nj}\beta + k_j + \varepsilon_{nj}$ with $E(\varepsilon_{nj}) = 0$. It is reasonable, therefore, to include a constant in V_{nj} for each alternative. However, since only differences in utility matter, only differences in the alternative-specific constants are relevant, not their absolute levels. To reflect this fact, the researcher must set the overall level of these constants.

The concept is readily apparent in the car-bus example. A specifi-

cation of utility that takes the form:

$$\begin{aligned} U_c &= \alpha T_c + \beta M_c + k_c^0 + \varepsilon_c \\ U_b &= \alpha T_b + \beta M_b + k_b^0 + \varepsilon_b \end{aligned}$$

with $k_b^0 - k_c^0 = d$, is equivalent to a model with

$$\begin{aligned} U_c &= \alpha T_c + \beta M_c + k_c^1 + \varepsilon_c \\ U_b &= \alpha T_b + \beta M_b + k_b^1 + \varepsilon_b \end{aligned}$$

where the difference in the new constants is the same as the difference in the old constants, namely $k_b^1 - k_c^1 = d = k_b^0 - k_c^0$. Any model with the same difference in constants is equivalent. In terms of estimation, it is impossible to estimate both constants since an infinite number of values of the two constants (any values that have the same difference) result in the same choice probabilities.

To account for this fact, the researcher must normalize the absolute levels of the constants. The standard procedure is to normalize one of the constants to zero. For example, the researcher might normalize the constant for the car alternative to zero:

$$\begin{aligned} U_c &= \alpha T_c + \beta M_c + \varepsilon_c \\ U_b &= \alpha T_b + \beta M_b + k_b + \varepsilon_b \end{aligned}$$

Under this normalization, the value of k_b is d , which is the difference in the original (unnormalized) constants. The bus constant is interpreted as the average impact of unincluded factors on the utility of bus *relative* to car.

With J alternatives, at most $J - 1$ alternative-specific constants can enter the model, with one of the constants normalized to zero. It is irrelevant which constant is normalized to zero: the other constants are interpreted as being relative to whichever one is set to zero. The researcher could normalize to some value other than zero, of course; however, there would be no point in doing so, since normalizing to zero is easier (the constant is simply left out of the model) and has the same effect.

Socio-demographic variables

The same issue affects the way that socio-demographic variables enter a model. Attributes of the alternatives, such as the time and cost of

travel on different modes, generally vary over alternatives. However, attributes of the decision-maker do not vary over alternatives. They can only enter the model if they are specified in ways that create differences in utility over alternatives.

Consider for example the impact of a person's income on the decision of whether to take bus or car to work. It is reasonable to suppose that a person's utility is higher with higher income, whether the person takes bus or car. Utility is specified as:

$$\begin{aligned} U_c &= \alpha T_c + \beta M_c + \theta_c^0 Y + \varepsilon_c \\ U_b &= \alpha T_b + \beta M_b + \theta_b^0 Y + k_b + \varepsilon_b \end{aligned}$$

where Y is income and θ_c^0 and θ_b^0 capture the effect of changes in income on the utility of taking car and bus, respectively. We expect that $\theta_c^0 > 0$ and $\theta_b^0 > 0$ since greater income makes people happier no matter what mode they take. However, $\theta_c^0 \neq \theta_b^0$ since income probably has a different effect on the person depending on their mode of travel. Since only differences in utility matter, the absolute levels of θ_c^0 and θ_b^0 cannot be estimated, only their difference. To set the level, one of these parameters is normalized to zero. The model becomes:

$$\begin{aligned} U_c &= \alpha T_c + \beta M_c + \varepsilon_c \\ U_b &= \alpha T_b + \beta M_b + \theta_b Y + k_b + \varepsilon_b \end{aligned}$$

where $\theta_b = \theta_b^0 - \theta_c^0$ and is interpreted as the differential effect of income on the utility of bus compared to car. The value of θ_b can be either positive or negative.

Socio-demographic variables can enter utility in other ways. For example, cost is often divided by income:

$$\begin{aligned} U_c &= \alpha T_c + \beta M_c / Y + \varepsilon_c \\ U_b &= \alpha T_b + \beta M_b / Y + \theta_b Y + k_b + \varepsilon_b \end{aligned}$$

The coefficient of cost in this specification is β/Y . Since this coefficient decreases in Y , the model reflects the concept that cost becomes less important in a person's decision-making, relative to other issues, when income rises.

When socio-demographic variables are interacted with attributes of the alternatives, there is no need to normalize the coefficients. The socio-demographic variables affect the differences in utility through

their interaction with the attributes of the alternatives. The difference $U_c - U_b = \dots \beta(M_c - M_b)/Y \dots$ varies with income since costs differ over alternatives.

Number of independent error terms

As given by equation (2.2), the choice probabilities take the form

$$P_{ni} = \int_{\varepsilon} I(\varepsilon_{nj} - \varepsilon_{ni} > V_{nj} - V_{ni} \quad \forall j \neq i) f(\varepsilon_n) d\varepsilon_n.$$

This probability is a J dimensional integral over the density of the J error terms in $\varepsilon_n = \langle \varepsilon_{n1}, \dots, \varepsilon_{nJ} \rangle$. The dimension can be reduced, however, through recognizing that only differences in utility matter. With J errors (one for each alternative), there are $J - 1$ error differences. The choice probability can be expressed as a $J - 1$ dimensional integral over the density of these error differences:

$$\begin{aligned} P_{ni} &= \text{Prob}(U_{ni} > U_{nj} \quad \forall j \neq i) \\ &= \text{Prob}(\varepsilon_{nj} - \varepsilon_{ni} < V_{ni} - V_{nj} \quad \forall j \neq i) \\ &= \text{Prob}(\tilde{\varepsilon}_{nji} < V_{ni} - V_{nj} \quad \forall j \neq i) \\ &= \int I(\tilde{\varepsilon}_{nji} < V_{ni} - V_{nj} \quad \forall j \neq i) g(\tilde{\varepsilon}_{ni}) d\tilde{\varepsilon}_{ni} \end{aligned}$$

where $\tilde{\varepsilon}_{nji} = \varepsilon_{nj} - \varepsilon_{ni}$ is the difference in errors for alternatives i and j ; $\tilde{\varepsilon}_{ni} = \langle \tilde{\varepsilon}_{n1i}, \dots, \tilde{\varepsilon}_{nJi} \rangle$ is the $J - 1$ dimensional vector of error differences, with the $\dots$ over all alternatives except i ; and $g(\cdot)$ is the density of these error differences. Expressed in this way, the choice probability is a $J - 1$ dimensional integral.

The density of the error differences $g(\cdot)$, and the density of the original errors, $f(\cdot)$, are related in a particular way. Suppose a model is specified with an error for each alternative: $\varepsilon_n = \langle \varepsilon_{n1}, \dots, \varepsilon_{nJ} \rangle$ with density $f(\varepsilon_n)$. This model is equivalent to a model with $J - 1$ errors defined as $\tilde{\varepsilon}_{njk} = \varepsilon_{nj} - \varepsilon_{nk}$ for any k and density $g(\tilde{\varepsilon}_{nk})$ derived from $f(\varepsilon_n)$. For any $f(\varepsilon_n)$, the corresponding $g(\tilde{\varepsilon}_{nk})$ can be derived. However, since ε_n has more elements than $\tilde{\varepsilon}_{nk}$, there is an infinite number of densities for the J error terms that give the same density for the $J - 1$ error differences. Stated equivalently, any $g(\tilde{\varepsilon}_{nk})$ is consistent with an infinite number of different $f(\varepsilon_n)$'s. Since choice probabilities can always be expressed as depending only on $g(\tilde{\varepsilon}_{nk})$, one dimension

of the density of $f(\varepsilon_n)$ is not identified and must be normalized by the researcher.

The normalization of $f(\varepsilon_n)$ can be handled in various ways. For some models, such as logit, the distribution of the error terms is sufficiently restrictive that the normalization occurs automatically with the assumptions on the distribution. For other models, such as probit, identification is often obtained by specifying the model only in terms of error differences, that is, by parameterizing $g(\cdot)$ without reference to $f(\cdot)$. In all but the simplest models, the researcher needs to consider the fact that only the density of error differences affects the probabilities and therefore is identified. In discussing the various models in subsequent chapters, we will return to this issue and how to handle it.

2.5.2 The overall scale of utility is irrelevant

Just as adding a constant to the utility of all alternatives does not change the decision-maker's choice, neither does multiplying each alternative's utility by a constant. The alternative with the highest utility is the same no matter how utility is scaled. The model $U_{nj}^0 = V_{nj} + \varepsilon_{nj} \forall j$ is equivalent to $U_{nj}^1 = \lambda V_{nj} + \lambda \varepsilon_{nj} \forall j$ for any $\lambda > 0$. To account for this fact, the researcher must normalize the scale of utility.

The standard way to normalize the scale of utility is to normalize the variance of the error terms. The scale of utility and the variance of the error terms are definitionally linked. When utility is multiplied by λ , the variance of each ε_{nj} changes by λ^2 : $Var(\lambda \varepsilon_{nj}) = \lambda^2 Var(\varepsilon_{nj})$. Therefore normalizing the variance of the error terms is equivalent to normalizing the scale of utility.

Normalization with iid errors

If the error terms are assumed to be independently, identically distributed, then the normalization for scale is straightforward. The researcher normalizes the error variance to some number, which is usually chosen for convenience. Since all the errors have the same variance by assumption, normalizing the variance of any of them sets the variance for them all.

When the observed portion of utility is linear in parameters, the normalization provides a way of interpreting coefficients. Consider the model $U_{nj}^0 = x'_{nj}\beta + \varepsilon_{nj}^0$ where the variance of the error terms is $Var(\varepsilon_{nj}^0) = \sigma^2$. Suppose the research normalizes the scale by setting

the error variance to 1. The original model becomes the following equivalent specification: $U_{nj}^1 = x'_{nj}(\beta/\sigma) + \varepsilon_{nj}^1$ with $Var(\varepsilon_{nj}^1) = 1$. The original coefficients β are divided by the standard deviation of the unobserved portion of utility. The new coefficients β/σ reflect, therefore, the impact of the observed variables *relative* to the standard deviation of the unobserved factors.

The same concepts apply for whatever number the researcher chooses for normalization. As we will see in the next chapter, the error variances in a standard logit model are traditionally normalized to $\pi^2/6$, which is about 1.6. In this case, the above model becomes $U_{nj} = x'_{nj}(\beta/\sigma)\sqrt{1.6} + \varepsilon_{nj}$ with $Var(\varepsilon_{nj}) = 1.6$. The coefficients still reflect the variance of the unobserved portion of utility. The only difference is that the coefficients are larger by a factor of $\sqrt{1.6}$.

While it is immaterial which number is used by the researcher for normalization, interpretation of model results must take the normalization into consideration. Suppose, for example, that a logit and an independent probit model were both estimated on the same data. As stated above, the error variance is normalized to 1.6 for logit. Suppose the researcher normalized the probit to have error variances of 1, which is traditional with independent probits. This difference in normalization must be kept in mind when comparing estimates from the two models. In particular, the coefficients in the logit model will be $\sqrt{1.6}$ times larger than those for the probit model, simply due to the difference in normalization. If the researcher does not take this scale difference into account when comparing the models, she might inadvertently think that the logit model implies that people care more about the attributes (since the coefficients are larger) than implied by the probit model. For example, in a mode choice model, supposed the estimated cost coefficient is -0.55 from a logit model and -0.45 from an independent probit model. It is incorrect to say that the logit model implies more sensitivity to costs than the probit model. The coefficients in one of the models must be adjusted to account for the difference in scale. The logit coefficients can be divided by $\sqrt{1.6}$, so that the error variance is 1, just like the probit model. With this adjustment, the comparable coefficients are -0.43 for the logit model and -0.45 for the probit model. The logit model implies less price sensitivity than the probit. Instead, the probit coefficients could be converted to the scale of the logit coefficients by multiplying them by $\sqrt{1.6}$, in which case the comparable coefficients would be -0.55 for logit and

−0.57 for probit.

A similar issue of interpretation arises when the same model is estimated on different data sets. The relative scale of the estimates from the two data sets reflects the relative variance of unobserved factors in the data sets. Suppose mode choice models were estimated in Chicago and Boston. For Chicago, the estimated cost coefficient is −0.55 and the estimated coefficient of time is −1.78. For Boston, the estimates are −0.81 and −2.69. The ratio of the cost coefficient to the time coefficient is very similar in the two cities: 0.309 in the Chicago and 0.301 in Boston. However, the scale of the coefficients is about fifty percent higher for Boston than Chicago. This scale difference means that the unobserved portion of utility has less variance in Boston than Chicago: since the coefficients are divided by the standard deviation of the unobserved portion of utility, lower coefficients mean higher standard deviation and hence variance. The models are revealing that factors other than time and cost have less effect on people in Boston than in Chicago. Stated more intuitively, time and cost have more importance, relative to unobserved factors, in Boston than in Chicago – which is consistent with the larger scale of the coefficients for Boston.

Normalization with heteroskedastic errors

In some situations, the variance of the error terms can be different for different segments of the population. The researcher cannot set the overall level of utility by normalizing the variance of the errors for all segments, since the variance is different in different segments. Instead, the researcher sets the overall scale of utility by normalizing the variance for one segment, and then estimates the variance (and hence scale) for each segment relative to this one segment.

For example, consider the situation described in the previous section, where the unobserved factors have greater variance in Chicago than in Boston. If separate models are estimated for Chicago and Boston, then the variance of the error term is normalized separately for each model. The scale of the parameters in each model reflect the variance of unincluded factors in that area. Suppose, however, that the researcher wants to estimate a model on data for both Chicago and Boston. She cannot normalize the variance of the unobserved factors for all travelers to the same number, since the variance is different for travelers in Boston than those in Chicago. Instead, the researcher sets

the overall scale of utility by normalizing the variance in one area (say Boston) and then estimates the variance in the other area *relative* that in the first area (the variance in Chicago relative to that in Boston).

The model in its original form is

$$\begin{aligned} U_{nj} &= \alpha T_{nj} + \beta M_{nj} + \varepsilon_{nj}^B \quad \forall n \text{ in Boston} \\ U_{nj} &= \alpha T_{nj} + \beta M_{nj} + \varepsilon_{nj}^C \quad \forall n \text{ in Chicago,} \end{aligned}$$

where the variance of ε_{nj}^B is not the same as the variance of ε_{nj}^C . Label the ratio of variances as $k = \text{Var}(\varepsilon_{nj}^C)/\text{Var}(\varepsilon_{nj}^B)$. We can divide the utility for travelers in Chicago by $\sqrt{k}$; this division doesn't affect their choices, of course, since the scale of utility doesn't matter. However, doing so allows us to re-write the model as

$$\begin{aligned} U_{nj} &= \alpha T_{nj} + \beta M_{nj} + \varepsilon_{nj} \quad \forall n \text{ in Boston} \\ U_{nj} &= (\alpha/\sqrt{k})T_{nj} + (\beta/\sqrt{k})M_{nj} + \varepsilon_{nj} \quad \forall n \text{ in Chicago,} \end{aligned}$$

where now the variance of ε_{nj} is the same for all n in both cities (since $\text{Var}(\varepsilon_{nj}^C/\sqrt{k}) = (1/k)\text{Var}(\varepsilon_{nj}^C) = [\text{Var}(\varepsilon_{nj}^B)/\text{Var}(\varepsilon_{nj}^C)]\text{Var}(\varepsilon_{nj}^C) = \text{Var}(\varepsilon_{nj}^B)$). The scale of utility is set by normalizing the variance of ε_{nj} . The parameter k , which is often called the scale parameter, is estimated along with β and α . The estimated value of k tells the researcher the variance of unobserved factors in Chicago relative to that in Boston. For example, $\hat{k} = 1.2$ implies that the variance of unobserved factors is twenty percent greater in Chicago than Boston.

The variance of the error term can differ over geographic regions, data sets, time, or other factors. In all cases, the researcher sets the overall scale of utility by normalizing one of the variances and then estimating the other variances relative to the normalized one. Swait and Louviere (1993) discuss the role of the scale parameter in discrete choice models, describing the variety of reasons that variances can differ over observations. As well as the traditional concept of variance in unobserved factors, psychological factors can come into play, depending on the choice situation and the interpretation of the researcher. For example, Bradley and Daly (1994) allow the scale parameter to vary over stated preference experiments in order to account for respondents' fatigue in answering the survey questions. Ben-Akiva and Morikawa (1990) allow the scale parameter to differ for respondents' stated intentions versus their actual market choices.

Normalization with correlated errors

In the discussion so far we have assumed that ε_{nj} is independent over alternatives. When the errors are correlated over alternatives, normalizing for scale is more complex. We have talked in terms of setting the scale of utility. However, since only differences in utility matter, it is more appropriate to talk in terms of setting the scale of utility *differences*. However, when errors are correlated, normalizing the variance of the error for one alternative is not sufficient to set the scale of utility differences.

The issue is most readily described in terms of a 4-alternative example. The utility for the four alternatives is $U_{nj} = V_{nj} + \varepsilon_{nj}$, $j = 1, \dots, 4$. The error vector $\varepsilon_n = \langle \varepsilon_{n1}, \dots, \varepsilon_{n4} \rangle$ has zero mean and covariance matrix:

$$\Omega = \begin{pmatrix} \sigma_{11} & \sigma_{12} & \sigma_{13} & \sigma_{14} \\ \cdot & \sigma_{22} & \sigma_{23} & \sigma_{24} \\ \cdot & \cdot & \sigma_{33} & \sigma_{34} \\ \cdot & \cdot & \cdot & \sigma_{44} \end{pmatrix} \quad (2.3)$$

where the dots refer to the corresponding elements on the upper part of the symmetric matrix.

Since only differences in utility matter, this model is equivalent to one in which all utilities are differenced from, say, the first alternative. The equivalent model is $\tilde{U}_{nj1} = \tilde{V}_{nj1} - \tilde{\varepsilon}_{nj1}$ for $j = 2, 3, 4$, where $\tilde{U}_{nj1} = U_{nj} - U_{n1}$, $\tilde{V}_{nj1} = V_{nj} - V_{n1}$, and the vector of error differences is $\tilde{\varepsilon}_{n1} = \langle (\varepsilon_{n2} - \varepsilon_{n1}), (\varepsilon_{n3} - \varepsilon_{n1}), (\varepsilon_{n4} - \varepsilon_{n1}) \rangle$. The variance of each error difference depends on the variances and covariances of the original errors. For example, the variance of the difference between the first and second errors is: $Var(\tilde{\varepsilon}_{n21}) = Var(\varepsilon_{n2} - \varepsilon_{n1}) = Var(\varepsilon_{n1}) + Var(\varepsilon_{n2}) - 2Cov(\varepsilon_{n1}, \varepsilon_{n2}) = \sigma_{11} + \sigma_{22} - 2\sigma_{12}$. We can similarly calculate the covariance between $\tilde{\varepsilon}_{n21}$, which is the difference between the first and second errors, and $\tilde{\varepsilon}_{n31}$, which is the difference between the first and third errors: $Cov(\tilde{\varepsilon}_{n21}, \tilde{\varepsilon}_{n31}) = E(\varepsilon_{n2} - \varepsilon_{n1})(\varepsilon_{n3} - \varepsilon_{n1}) = E(\varepsilon_{n2}\varepsilon_{n3} - \varepsilon_{n2}\varepsilon_{n1} - \varepsilon_{n3}\varepsilon_{n1} + \varepsilon_{n1}\varepsilon_{n1}) = \sigma_{23} - \sigma_{21} - \sigma_{31} + \sigma_{11}$. The covariance matrix for the vector of error differences becomes:

$$\tilde{\Omega}_1 = \begin{pmatrix} \sigma_{11} + \sigma_{22} - 2\sigma_{12} & \sigma_{11} + \sigma_{23} - \sigma_{12} - \sigma_{13} & \sigma_{11} + \sigma_{24} - \sigma_{12} - \sigma_{14} \\ \cdot & \sigma_{11} + \sigma_{33} - 2\sigma_{13} & \sigma_{11} + \sigma_{34} - \sigma_{13} - \sigma_{14} \\ \cdot & \cdot & \sigma_{11} + \sigma_{44} - 2\sigma_{14} \end{pmatrix}.$$

Setting the variance of one of the original errors is not sufficient to set the variance of the error differences. For example, if the variance for

the first alternative were set to some number $\sigma_{11} = k$, the variance of the difference between the errors for the first two alternatives becomes $k + \sigma_{22} - \sigma_{12}$. An infinite number of values for $\sigma_{22} - \sigma_{12}$ provide equivalent models.

A common way to set the scale of utility when errors are not iid is to normalize the variance of one of the error differences to some number. Setting the variance of an error difference sets the scale of utility differences and hence of utility. Suppose we normalize the variance of $\tilde{\varepsilon}_{n21}$ to 1. The covariance matrix for the error differences, expressed in terms of the covariances of the original errors, becomes:

$$\begin{pmatrix} 1 & (\sigma_{11} + \sigma_{23} - \sigma_{12} - \sigma_{13})/m & (\sigma_{11} + \sigma_{24} - \sigma_{12} - \sigma_{14})/m \\ \cdot & (\sigma_{11} + \sigma_{33} - 2\sigma_{13})/m & (\sigma_{11} + \sigma_{34} - \sigma_{13} - \sigma_{14})/m \\ \cdot & \cdot & (\sigma_{11} + \sigma_{44} - 2\sigma_{14})/m \end{pmatrix}, \quad (2.4)$$

where $m = \sigma_{11} + \sigma_{22} - 2\sigma_{12}$. Utility is divided by $\sqrt{\sigma_{11} + \sigma_{22} - 2\sigma_{12}}$ to obtain this scaling.

Note that when the error terms are iid, normalizing the variance of one of these errors automatically normalizes the variance of the error differences. With iid errors, $\sigma_{jj} = \sigma_{ii}$ and $\sigma_{ij} = 0$ for $i \neq j$. Therefore, if σ_{11} is normalized to k , then the variance of the error difference becomes $\sigma_{11} + \sigma_{22} - 2\sigma_{12} = k + k - 0 = 2k$. The variance of the error difference is indeed being normalized, the same as with non-iid errors.

Normalization has implications for the number of parameters that can be estimated in the covariance matrix. The covariance of the original errors, Ω in equation (2.3), has ten elements in our 4-alternative example. However, the covariance matrix of the error differences has six elements, one of which is normalized to set the scale of utility differences. The covariance matrix for error differences with the variance of the first error difference normalized to k takes the form:

$$\tilde{\Omega}_1^* = \begin{pmatrix} k & \omega_{ab} & \omega_{ac} \\ \cdot & \omega_{bb} & \omega_{bc} \\ \cdot & \cdot & \omega_{cc} \end{pmatrix} \quad (2.5)$$

which has only five parameters. By recognizing the fact that only differences matter and that the scale of utility is arbitrary, the number of covariance parameters drops from ten to five. A model with J alternatives has at most $J(J-1)/2 - 1$ covariance parameters after normalization.

Interpretation of the model is affected by the normalization. Suppose for example that the elements of matrix (2.5) were estimated. The parameter ω_{bb} is the variance of the difference between the errors for the first and third alternatives *relative* to the variance of the difference between the errors for the first and second alternatives. Complicating interpretation even further is the fact that the variance of the difference between the errors for two alternatives reflects the variances of both as well as their covariance.

As we will see, the normalization of logit and nested logit models is automatic with the distributional assumptions that are placed on the error terms. Interpretation under these assumptions is relatively straightforward. For mixed logit and probit, fewer assumptions are placed on the distribution of error terms such that normalization is not automatic. The researcher must keep the normalization issues in mind when specifying and interpreting a model. We return to this topic when discussing each discrete choice model in subsequent chapters.

2.6 Aggregation

Discrete choice models operate at the level of individual decision-makers. However, the researcher is usually interested in some aggregate measure, such as the average probability within a population or the average response to a change in some factor.

In linear regression models, estimates of aggregate values of the dependent variable are obtained by inserting aggregate values of the explanatory variables into the model. For example, suppose h_n is housing expenditures of person n , y_n is the income of the person, and the model relating them is $h_n = \alpha + \beta y_n$. Since this model is linear, the average expenditure on housing is simply calculated as $\alpha + \beta \bar{y}$, where $\bar{y}$ is average income. Similarly, the average response to a one-unit change in income is simply β , since β is the response for each person.

Discrete choice models are not linear in explanatory variables, and, consequently, inserting aggregate values of the explanatory variables into the models will not provide an unbiased estimate of the average probability or average response. The point can be made visually. Consider Figure 2.1, which gives the probabilities of choosing a particular alternative for two individuals with the observed portion of their utility (their “representative utility”) being a and b . The average probability is the average of the probabilities for the two people, namely,

$(P_a + P_b)/2$. The average representative utility is $(a + b)/2$, and the probability evaluated at this average is the point on the curve above $(a + b)/2$. As shown for this case, the average probability is greater than the probability evaluated at the average representative utility. In general, the probability evaluated at the average representative utility underestimates the average probability when the individuals' choice probabilities are low and overestimates when they are high.

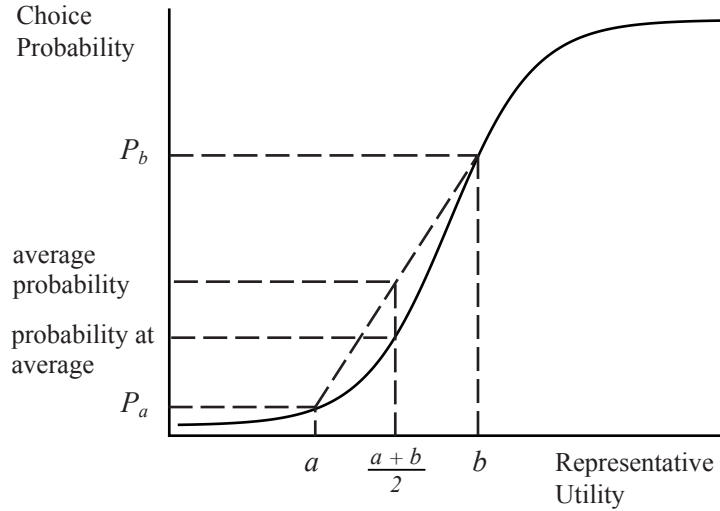


Figure 2.1: Difference between average probability and probability calculated at average representative utility.

Estimating the average response by calculating derivatives and elasticities at the average of the explanatory variables is similarly problematic. Consider Figure 2.2, depicting two individuals with representative utilities a and b . The derivative of the choice probability for a change in representative utility is small for both of these people (the slope of the curve above a and b). Consequently, the average derivative is also small. However, the derivative at the average representative utility is very high (the slope above $(a + b)/2$.) Estimating average response in this way can be seriously misleading. In fact, Talvitie (1976) found, in a mode choice situation, that elasticities at the average representative utility can be as much as two or three times greater or less than the average of the individual elasticities.

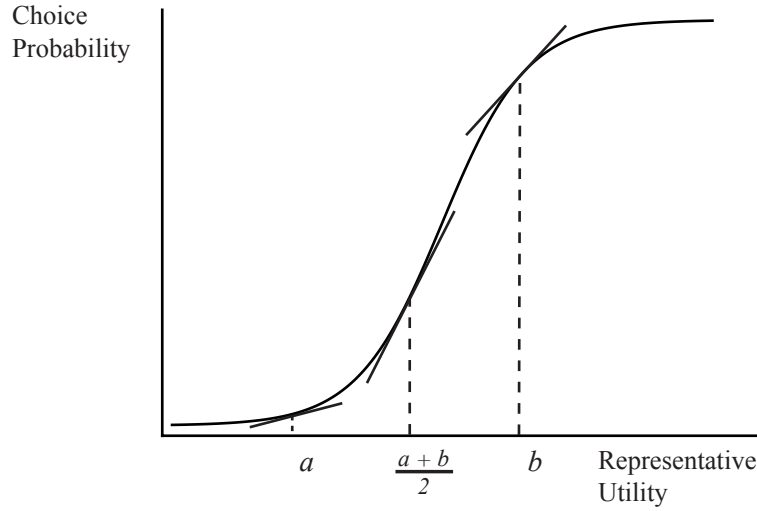


Figure 2.2: Difference between average response and response calculated at average representative utility.

Aggregate outcome variables can be obtained consistently from discrete choice models in two ways, by sample enumeration or segmentation. We discuss each approach below.

2.6.1 Sample enumeration

The most straightforward, and by far the most popular, approach is sample enumeration, by which the choice probabilities of each decision-maker in a sample are summed, or averaged, over decision-makers. Consider a discrete choice model that gives probability P_{ni} that decision-maker n will choose alternative i from a set of alternatives. Suppose a sample of N decision-makers, labeled $n = 1, \dots, N$, is drawn from the population for which aggregate statistics are required. (This sample might be the sample on which the model was estimated. However, it could also be a different sample, collected in a different area or a later date than the estimation sample.) Each sampled decision-maker n has some weight associated with him, w_n , representing the number of decision-makers similar to him in the population. For samples based on exogenous factors, this weight is the inverse of the probability that the decision-maker was selected into the sample. If the sample is purely

random, then w_n is the same for all n ; and if the sample is stratified random, then w_n is the same for all n within a stratum.

A consistent estimate of the total number of decision-makers in the population who choose alternative i , labeled $\hat{N}_i$, is simply the weighted sum of the individual probabilities:

$$\hat{N}_i = \sum_n w_n P_{ni}.$$

The average probability, which is the estimated market share, is $\hat{N}_i/N$. Average derivatives and elasticities are similarly obtained by calculating the derivative and elasticity for each sampled person and taking the weighted average.

2.6.2 Segmentation

When the number of explanatory variables is small, and those variables take only a few values, it is possible to estimate aggregate outcomes without utilizing a sample of decision-makers. Consider, for example, a model with only two variables entering the representative utility of each alternative: education level and gender. Suppose the education variable consists of four categories: did not complete high school, completed high school but did not attend college, attended college but did not receive a degree, received a college degree. Then the total number of different types of decision-makers (called "segments") is eight: the four education levels for each of the two genders. Choice probabilities vary only over these eight segments, not over individuals within each segment.

If the researcher has data on the number of people in each segment, then the aggregate outcome variables can be estimated by calculating the choice probability for each segment and taking the weighted sum of these probabilities. The number of people estimated to choose alternative i is

$$\hat{N}_i = \sum_{s=1}^8 w_s P_{si},$$

where P_{si} is the probability that a decision-maker in segment s chooses alternative i , and w_s is the number of decision-makers in segment s .

2.7 Forecasting

For forecasting into some future year, the procedures described above for aggregate variables are applied. However, the exogenous variables and/or the weights are adjusted to reflect changes that are anticipated over time. With sample enumeration, the sample is adjusted so that it *looks like* a sample that would be drawn in the future year. For example, to forecast the number of people who will choose a given alternative five years in the future, a sample drawn from the current year is adjusted to reflect changes in socioeconomic and other factors that are expected to occur over the next five years. The sample is adjusted by (i) changing the value of the variables associated with each sampled decision-maker (e.g., increasing each decision-maker's income to represent real income growth over time), and/or (ii) changing the weight attached to each decision-maker to reflect changes over time in the number of decision-makers in the population that are similar to the sampled decision-maker (e.g., increasing the weights for one-person households and decreasing weights for large households to reflect expected decreases in household size over time.)

For the segmentation approach, changes in explanatory variables over time are represented by changes in the number of decision-makers in each segment. The explanatory variables themselves cannot logically be adjusted since the distinct values of the explanatory variables define the segments. Changing the variables associated with a decision-maker in one segment simply shifts the decision-maker to another segment.

2.8 Recalibration of constants

As described in section (2.5.1), alternative-specific constants are often included in a model to capture the average impact of unobserved factors. In forecasting, it is often useful to adjust these constants, to reflect the fact that unobserved factors are different for the forecast area or year compared to the estimation sample. Market share data from the forecast area can be used to “recalibrate” the constants appropriately. The recalibrated model can then be used to predict changes in market shares due to changes in explanatory factors.

An iterative process is used to recalibrate the constants. Let α_j^0 be the estimated alternative-specific constant for alternative j . The superscript 0 is used to indicate that these are the starting values in

the iterative process. Let S_i denote the share of decision-makers in the forecast area that choose alternative j in the “base” year (usually, the latest year for which such data are available.) Using the discrete choice model with its original values of $\alpha_j^0 \forall j$, predict the share of decision-makers in the forecast area who will choose each alternative. Label these predictions $\hat{S}_j^0 \forall j$. Compare the predicted shares with the actual shares. If the actual share for an alternative exceeds the predicted share, raise the constant for that alternative. Lower the constant if the actual share is below the predicted. An effective adjustment is:

$$\alpha_j^1 = \alpha_j^0 + \ln(S_j/\hat{S}_j^0).$$

With the new constants, predict the share again, compare with the actual shares, and if needed adjust the constants again. The process is repeated until the forecasted shares are sufficiently close to the actual shares. The model with these recalibrated constants is then used to predict changes from base-year shares due to changes in observed factors that affect decision-makers’ choices.

Chapter 3

Logit

3.1 Choice probabilities

By far the easiest and most widely used discrete choice model is logit. Its popularity is due to the fact that the formula for the choice probabilities takes a closed form and is readily interpretable. Originally, the logit formula was derived by Luce (1959) from assumptions about the characteristics of choice probabilities, namely the “independence from irrelevant alternatives” property discussed in 3.3.2. Marschak (1960) showed that these axioms implied that the model is consistent with utility maximization. The relation of the logit formula to the distribution of unobserved utility (as opposed to the characteristics of choice probabilities) was developed by Marley, as cited by Luce and Suppes (1965), who showed that the extreme value distribution leads to the logit formula. McFadden (1974) completed the analysis by showing the opposite: that the logit formula for the choice probabilities necessarily implies that unobserved utility is distributed extreme value. In his Nobel lecture, McFadden (2001) provides a fascinating history of the development of this path-breaking model.

To derive the logit model, we use the general notation from Chapter 2 and add a specific distribution for unobserved utility. A decision-maker, labeled n , faces J alternatives. The utility that the decision-maker obtains from alternative j is decomposed into (1) a part labeled V_{nj} that is known by the researcher up to some parameters, and (2) an unknown part ε_{nj} that is treated by the researcher as random: $U_{nj} = V_{nj} + \varepsilon_{nj} \forall j$. The logit model is obtained by assuming that each ε_{nj} is distributed independently, identically extreme value. The distri-

bution is also called Gumbel and type I extreme value (and sometimes, mistakenly, Weibull.) The density for each unobserved component of utility is

$$f(\varepsilon_{nj}) = e^{-\varepsilon_{nj}} e^{-e^{-\varepsilon_{nj}}} \quad (3.1)$$

and the cumulative distribution is

$$F(\varepsilon_{nj}) = e^{-e^{-\varepsilon_{nj}}}. \quad (3.2)$$

The variance of this distribution is $\pi^2/6$. By assuming the variance is $\pi^2/6$, we are implicitly normalizing the scale of utility, as discussed in section 2.5. We return to this issue, and its impact on interpretation, in the next section. The mean of the extreme value distribution is not zero; however, the mean is immaterial since only differences in utility matter (see Chapter 2) and the difference between two random terms that have the same mean has itself a mean of zero.

The difference between two extreme value variables is distributed logistic. That is, if ε_{nj} and ε_{ni} are iid extreme value, then $\varepsilon_{nji}^* = \varepsilon_{nj} - \varepsilon_{ni}$ follows the logistic distribution

$$F(\varepsilon_{nji}^*) = \frac{e^{\varepsilon_{nji}^*}}{1 + e^{\varepsilon_{nji}^*}}. \quad (3.3)$$

This formula is sometimes used in describing binary logit models, i.e., models with two alternatives. Using the extreme value distribution for the errors (and hence the logistic distribution for the error differences) is nearly the same as assuming that the errors are independently normal. The extreme value gives slightly fatter tails than a normal, which means that it allows for slightly more aberrant behavior than the normal. Usually, however, the difference between extreme value and independent normal errors is indistinguishable empirically.

The key assumption is not so much the shape of the distribution as that the errors are independent of each other. This independence means that the unobserved portion of utility for one alternative is unrelated to the unobserved portion of utility for another alternative. It is a fairly restrictive assumption, and the development of other models such as those described in Chapters 4-6 has arisen largely for the purpose of avoiding this assumption and allowing for correlated errors.

It is important to realize that the independence assumption is not as restrictive as it might at first seem, and in fact can be interpreted

as a natural outcome of a well-specified model. Recall from Chapter 2 that ε_{nj} is defined as the difference between the utility that the decision-maker actually obtains, U_{nj} , and the representation of utility that the researcher has developed using observed variables, V_{nj} . As such, ε_{nj} and its distribution depend on the researcher's specification of representative utility; it is not defined by the choice situation *per se*. In this light, the assumption of independence obtains a different stature. Under independence, the error for one alternative provides no information to the researcher about the error for another alternative. Stated equivalently, the researcher has specified V_{nj} sufficiently that the remaining, unobserved portion of utility is essentially "white noise." In a deep sense, the ultimate goal of the researcher is to represent utility so well that the only remaining aspects constitute simply white noise. That is: the goal is to specify utility well enough that a logit model is appropriate. Seen in this way, the logit model is the ideal rather than a restriction.

If the researcher thinks that the unobserved portion of utility is correlated over alternatives given his specification of representative utility, then she has three options (1) use a different model that allows for correlated errors, such as those described in Chapters 4-6, (2) re-specify representative utility so that the source of the correlation is captured explicitly such that the remaining errors are independent, or (3) use the logit model under the current specification of representative utility considering the model to be an approximation. The viability of the last option depends, of course, on the goals of the research. Violations of the logit assumptions seem to have less impact when estimating average preferences than when forecasting substitution patterns. These issues are discussed in subsequent sections.

We now derive the logit choice probabilities, following McFadden (1974). The probability that decision-maker n chooses alternative i is

$$\begin{aligned} P_{ni} &= \text{Prob}(V_{ni} + \varepsilon_{ni} > V_{nj} + \varepsilon_{nj} \quad \forall j \neq i) \\ &= \text{Prob}(\varepsilon_{nj} < \varepsilon_{ni} + V_{ni} - V_{nj} \quad \forall j \neq i) \end{aligned} \quad (3.4)$$

If ε_{ni} is considered given, this expression is the cumulative distribution for each ε_{nj} evaluated at $\varepsilon_{ni} + V_{ni} - V_{nj}$, which, according to (3.2), is $\exp(-\exp - (\varepsilon_{ni} + V_{ni} - V_{nj}))$. Since the ε 's are independent, this cumulative distribution over all $j \neq i$ is the product of the individual

cumulative distributions:

$$P_{ni} \mid \varepsilon_{ni} = \prod_{j \neq i} e^{-e^{-(\varepsilon_{ni} + V_{ni} - V_{nj})}}.$$

Of course, ε_{ni} is not given, and so the choice probability is the integral of $P_{ni} \mid \varepsilon_{ni}$ over all values of ε_{ni} weighted by its density (3.1):

$$P_{ni} = \int \left(\prod_{j \neq i} e^{-e^{-(\varepsilon_{ni} + V_{ni} - V_{nj})}} \right) e^{-\varepsilon_{ni}} e^{-e^{-\varepsilon_{ni}}} d\varepsilon_{ni}. \quad (3.5)$$

Some algebraic manipulation of this integral results in a succinct, closed-form expression:

$$P_{ni} = \frac{e^{V_{ni}}}{\sum_j e^{V_{nj}}} \quad (3.6)$$

which is the logit choice probability. The algebra that obtains (3.6) from (3.5) is given in the last section of this chapter, for readers who are interested.

Representative utility is usually specified to be linear in parameters: $V_{nj} = \beta' x_{nj}$ where x_{nj} is a vector of observed variables relating to alternative j . With this specification, the logit probabilities become:

$$P_{ni} = \frac{e^{\beta' x_{ni}}}{\sum_j e^{\beta' x_{nj}}}.$$

Under fairly general conditions, any function can be approximated arbitrarily closely by one that is linear in parameters. The assumption is therefore fairly benign. Importantly, McFadden (1974) demonstrated that the log-likelihood function with these choice probabilities is globally concave in parameters β , which helps in the numerical maximization procedures (as discussed in Chapter 8.) Numerous computer packages contain routines for estimation of logit models with linear-in-parameters representative utility.

The logit probabilities exhibit several desirable properties. First, P_{ni} is necessarily between zero and one, as required for a probability. When V_{ni} rises, reflecting an improvement in the observed attributes of the alternative, with $V_{nj} \forall j \neq i$ held constant, P_{ni} approaches one. And P_{ni} approaches zero when V_{ni} decreases, since the exponential in the numerator of (3.6) approaches zero as V_{ni} approaches $-\infty$. The logit probability for an alternative is never exactly zero. If the research

believes that an alternative has actually no chance of being chosen by a decision-maker, the researcher can exclude that alternative from the choice set. A probability of exactly 1 is obtained only if the choice set consists of a single alternative.

Second, the choice probabilities for all alternatives sum to one: $\sum_{i=1}^J P_{ni} = \sum_i \exp(V_{ni}) / \sum_j \exp(V_{nj}) = 1$. The decision-maker necessarily chooses one of the alternatives. The denominator in (3.6) is simply the sum of the numerator over all alternatives, which gives this summing-up property automatically. With logit, as well as with some more complex models such as the nested logit models of Chapter 4, interpretation of the choice probabilities is facilitated by recognition that the denominator serves to assure that the probabilities sum to one. In other models, such as mixed logit and probit, there is no denominator *per se* to interpret in this way.

The relation of the logit probability to representative utility is sigmoid, or S-shaped, as shown in Figure 3.1. This shape has implications for the impact of changes in explanatory variables. If the representative utility of an alternative is very low compared with other alternatives, a small increase in the utility of the alternative has little effect on the probability of its being chosen: the other alternatives are still sufficiently better that this small improvement doesn't help much. Similarly, if one alternative is far superior to the others in observed attributes, a further increase in its representative utility has little effect on the choice probability. The point at which the increase in representative utility has the greatest effect on the probability of its being chosen is when the probability is close to 0.5, meaning a 50-50 chance of the alternative being chosen. In this case, a small improvement "tips the balance" in people's choices, inducing a large change in probability. The sigmoid shape of logit probabilities is shared by most discrete choice models and has important implications for policy makers. For example, improving bus service in areas where the service is so poor that few travelers take the bus would be less effective, in terms of transit ridership, than making the same improvement in areas where bus service is already sufficiently good to induce a moderate share of travelers to choose it (but not so good that nearly everyone does.)

The logit probability formula is easily interpretable in the context of an example. Consider a binary choice situation first: a household's choice between a gas and electric heating system. Suppose that the

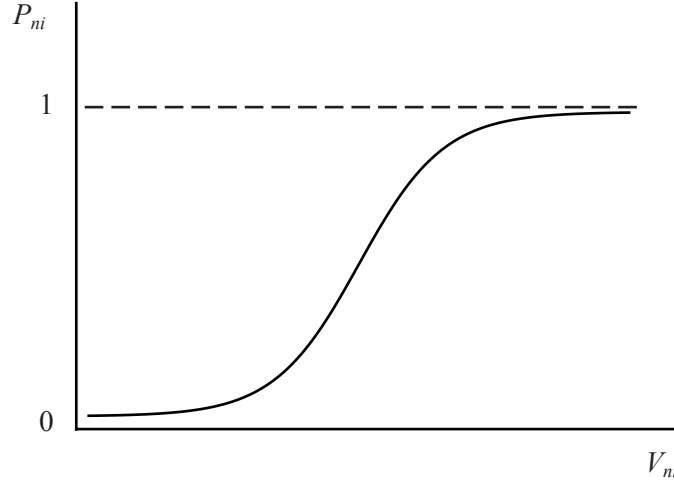


Figure 3.1: Graph of logit curve.

utility the household obtains from each type of system depends only on the purchase price, annual operating cost, and the household's view of the convenience and quality of heating with each type of system and the relative aesthetics of the systems within the house. The first two of these factors can be observed by the researcher, but the researcher cannot observe the others. If the researcher considers the observed part of utility to be a linear function of the observed factors, then the utility of each heating system can be written as: $U_g = \beta_1 PP_g + \beta_2 OC_g + \varepsilon_g$ and $U_e = \beta_1 PP_e + \beta_2 OC_e + \varepsilon_e$, where the subscripts g and e denote gas and electric, PP and OC are the purchase price and operating cost, β_1 and β_2 are scalar parameters, and the subscript n for the household is suppressed. Since higher costs mean less money to spend on other goods, we expect utility to drop as purchase price or operating cost rises (with all else held constant): $\beta_1 < 0$ and $\beta_2 < 0$.

The unobserved component of utility for each alternative, ε_g and ε_e , varies over households depending on how each household views the quality, convenience and aesthetics of each type of system. If these unobserved components are distributed iid extreme value, then the probability that the household will choose gas heating is

$$P_g = \frac{e^{\beta_1 PP_g + \beta_2 OC_g}}{e^{\beta_1 PP_g + \beta_2 OC_g} + e^{\beta_1 PP_e + \beta_2 OC_e}} \quad (3.7)$$

and the probability of electric heating is the same but with $\exp(\beta_1 PP_e + \beta_2 OC_e)$ as the numerator. The probability of choosing a gas system decreases if its purchase price or operating cost rises while that of the electric system remains the same (assuming that β_1 and β_2 are negative as expected.)

As in most discrete choice models, the ratio of coefficients in this example has economic meaning. In particular, the ratio β_2/β_1 represents the household's willingness to pay for operating cost reductions. If β_1 were estimated as -0.20 and β_2 as -1.14 , these estimates would imply that households are willing to pay up to $-1.14 / -0.20 = 5.70$ dollars more for a system whose annual operating costs are one dollar less. This relation is derived as follows. By definition, a household's willingness to pay for operating cost reductions is the increase in purchase price that keeps the household's utility constant given a reduction in operating costs. We take the total derivative of utility with respect to purchase price and operating cost and set this derivative to zero so that utility doesn't change: $\partial U = \beta_1 \partial PP + \beta_2 \partial OC = 0$. We then solve for the change in purchase price that keeps utility constant (i.e., satisfies this equation) for a change in operating costs: $\partial PP / \partial OC = -\beta_2 / \beta_1$. The negative sign indicates that the two changes are in the opposite direction: to keep utility constant, purchase price rises when operating cost decreases.

In this binary choice situation, the choice probabilities can be expressed in another, even more succinct form. Dividing the numerator and denominator of (3.7) by the numerator, and recognizing that $\exp(a)/\exp(b) = \exp(a - b)$, we have

$$P_g = \frac{1}{1 + e^{(\beta_1 PP_e + \beta_2 OC_e) - (\beta_1 PP_g + \beta_2 OC_g)}}.$$

In general, binary logit probabilities with representative utilities V_{n1} and V_{n2} can be written $P_{n1} = 1/(1 + \exp(V_{n2} - V_{n1}))$ and $P_{n2} = 1/(1 + \exp(V_{n1} - V_{n2}))$. If only demographics of the decision-maker, s_n , enter the model, and the coefficients of these demographic variables are normalized to zero for the first alternative (as described in Chapter 2), the probability of the first alternative is $P_{n1} = 1/(1 + e^{\alpha' s_n})$, which is the form that is used in most textbooks and computer manuals for binary logit.

Multinomial choice is a simple extension. Suppose there is a third type of heating system, namely oil-fueled. The utility of the oil system

is specified as the same form as for the electric and gas systems: $U_o = \beta_1 PP_o + \beta_2 OC_o + \varepsilon_o$. With this extra option being available, the probability that the household chooses a gas system is:

$$P_g = \frac{e^{\beta_1 PP_g + \beta_2 OC_g}}{e^{\beta_1 PP_g + \beta_2 OC_g} + e^{\beta_1 PP_e + \beta_2 OC_e} + e^{\beta_1 PP_o + \beta_2 OC_o}}$$

which is the same as (3.7) except that an extra term is included in the denominator to represent the oil heater. Since the denominator is larger while the numerator is the same, the probability of choosing a gas system is smaller when an oil system is an option than when not, as one would expect in the real world.

3.2 The scale parameter

In the previous section we derived the logit formula under the assumption that the unobserved factors are distributed extreme value with variance $\pi^2/6$. Setting the variance to $\pi^2/6$ is equivalent to normalizing the model for the scale of utility, as discussed in section 2.5. It is useful to make these concepts more explicit, to show the role that the variance of the unobserved factors plays in logit models.

In general, utility can be expressed as $U_{nj}^* = V_{nj} + \varepsilon_{nj}^*$ where the unobserved portion has variance $\sigma^2 * (\pi^2/6)$. That is, the variance is any number, re-expressed as a multiple of $\pi^2/6$. Since the scale of utility is irrelevant to behavior, utility can be divided by σ without changing behavior. Utility becomes $U_{nj} = V_{nj}/\sigma + \varepsilon_{nj}$ where $\varepsilon_{nj} = \varepsilon_{nj}^*/\sigma$. Now the unobserved portion has variance $\pi^2/6$: $Var(\varepsilon_{nj}) = Var(\varepsilon_{nj}^*/\sigma) = (1/\sigma^2)Var(\varepsilon_{nj}^*) = (1/\sigma^2) * \sigma^2 * (\pi^2/6) = \pi^2/6$. The choice probability is

$$P_{ni} = \frac{e^{V_{ni}/\sigma}}{\sum_j e^{V_{nj}/\sigma}}$$

which is the same formula as in equation 3.6 but with representative utility divided by σ . If V_{nj} is linear in parameters with coefficient β^* , the choice probabilities become:

$$P_{ni} = \frac{e^{(\beta^*/\sigma)'x_{ni}}}{\sum_j e^{(\beta^*/\sigma)'x_{nj}}}$$

Each of the coefficients is scaled by $1/\sigma$. The parameter σ is called the scale parameter, because it scales the coefficients to reflect the variance of the unobserved portion of utility.

Only the ratio β^*/σ can be estimated; β^* and σ are not separately identified. Usually, the model is expressed in its scaled form, with $\beta = \beta^*/\sigma$, which gives the standard logit expression

$$P_{ni} = \frac{e^{\beta' x_{ni}}}{\sum_j e^{\beta' x_{nj}}}$$

The parameters β are estimated, but for interpretation it is useful to recognize that these estimated parameters are actually estimates of the “original” coefficients β^* divided by the scale parameter σ . The coefficients that are estimated indicate the impact of each observed variable *relative to* the variance of the unobserved factors. A larger variance in unobserved factors leads to lower coefficients, even if the observed factors have the same impact on utility (i.e., higher σ means lower β even if β^* is the same.)

The scale parameter does not affect the ratio of any two coefficients, since it drops out of the ratio; for example, $\beta_1/\beta_2 = (\beta_1^*/\sigma)/(\beta_2^*/\sigma) = \beta_1^*/\beta_2^*$, where the subscripts refer to the first and second coefficients. Willingness to pay, values of time, and other measures of marginal rates of substitution are not affected by the scale parameter. Only the interpretation of the magnitudes of all coefficients is affected.

So far we have assumed that the variance of the unobserved factors is the same for all decision-makers, since the same σ is used for all n . Suppose instead that the unobserved factors have greater variance for some decision-makers than others. In section 2.5, we discuss a situation where the variance of unobserved factors is different in Boston than Chicago. Denote the variance for all decision-makers in Boston as $(\sigma^B)^2(\pi^2/6)$ and that for decision-makers in Chicago as $(\sigma^C)^2(\pi^2/6)$. The ratio of variance in Chicago to that in Boston is $k = (\sigma^C/\sigma^B)^2$. The choice probabilities for people in Boston become

$$P_{ni} = \frac{e^{\beta' x_{ni}}}{\sum_j e^{\beta' x_{nj}}}$$

and for people in Chicago are

$$P_{ni} = \frac{e^{(\beta/\sqrt{k})' x_{ni}}}{\sum_j e^{(\beta/\sqrt{k})' x_{nj}}}$$

where $\beta = \beta^*/\sigma^B$. The ratio of variances k is estimated along with the coefficients β . The estimated β 's are interpreted as being relative

to the variance of unobserved factors in Boston, and the estimated k provides information on the variance in Chicago relative to that in Boston. More complex relations can be obtained by allowing the variance for an observation to depend on more factors. Also, data from different data sets can often be expected to have different variance for unobserved factors, giving a different scale parameter for each data set. Ben-Akiva and Morikawa (1990) and Swait and Louviere (1993) discuss these issues and provide more examples.

3.3 Power and limitations of logit

Three topics elucidate the power of logit models to represent choice behavior, as well as delineating the limits to that power. These topics are: taste variation, substitution patterns, and repeated choices over time. The applicability of logit models can be summarized as follows:

1. Logit can represent systematic taste variation (that is, taste variation that relates to observed characteristics of the decision-maker) but not random taste variation (differences in tastes that cannot be linked to observed characteristics).
2. The logit model implies proportional substitution across alternatives, given the researcher's specification of representative utility. To capture more flexible forms of substitution, other models are needed.
3. If unobserved factors are independent over time in repeated choice situations, then logit can capture the dynamics of repeated choice, including state-dependence. However, logit cannot handle situations where unobserved factors are correlated over time.

We elaborate each of these statements in the next three subsections.

3.3.1 Taste variation

The value or importance that decision-makers place on each attribute of the alternatives varies, in general, over decision-makers. For example, the size of a car is probably more important to households with many members than to smaller households. Low-income households are probably more concerned about the purchase price of a good, relative to its other characteristics, than higher-income households. In

choosing which neighborhood to live in, households with young children will be more concerned about the quality of schools than those without children. And so on. Decision-makers' tastes also vary for reasons that are not linked to observed demographic characteristics, just because different people are different. Two people who have the same income, education, etc., will make different choices, reflecting their individual preferences and concerns.

Logit models can capture taste variations, but only within limits. In particular, tastes that vary systematically with respect to observed variables can be incorporated in logit models, while tastes that vary with unobserved variables or purely randomly cannot be handled. The following example illustrates the distinction.

Consider households' choice among makes and models of cars to buy. Suppose for simplicity that the only two attributes of cars that the researcher observes are the purchase price, PP_j for make/model j , and inches of shoulder room, SR_j , which is a measure of the interior size of a car. The value that households place on these two attributes varies over households, and so utility is written as

$$U_{nj} = \alpha_n SR_j + \beta_n PP_j + \varepsilon_{nj}, \quad (3.8)$$

where α_n and β_n are parameters specific to household n .

The parameters vary over households reflecting differences in taste. Suppose for example that the value of shoulder room varies with the number of members in the households, M_n , but nothing else:

$$\alpha_n = \rho M_n,$$

so that as M_n increases, the value of shoulder room, α_n , also increases. Similarly, suppose the importance of purchase price is inversely related to income, I_n , so that low income households place more importance on purchase price:

$$\beta_n = \theta / I_n.$$

Substituting these relations into (3.8) produces

$$U_{nj} = \rho(M_n SR_j) + \theta(PP_j / I_n) + \varepsilon_{nj}.$$

Under the assumption that each ε_{nj} is iid extreme value, a standard logit model obtains with two variables entering representative utility, both of which are an interaction of a vehicle attribute with a household characteristic.

Other specifications for the variation in tastes can be substituted. For example, the value of shoulder room might be assumed to increase with household size, but at a decreasing rate, so that $\alpha_n = \rho M_n + \phi M_n^2$ where ρ is expected to be positive and ϕ negative. Then $U_{nj} = \rho(M_n SR_j) + \phi(M_n^2 SR_j) + \theta(PP_j/I_n) + \varepsilon_{nj}$, which results in a logit model with three variables entering representative utility.

The limitation of the logit model arises when we attempt to allow tastes to vary with respect to unobserved variables or purely randomly. Suppose for example that the value of shoulder room varied with household size plus some other factors (e.g., size of the people themselves, or frequency with which the household travels together) that are unobserved by the researcher and hence considered random:

$$\alpha_n = \rho M_n + \mu_n,$$

where μ_n is a random variable. Similarly, the importance of purchase price consists of its observed and unobserved components:

$$\beta_n = \theta/I_n + \eta_n.$$

Substituting into (3.8) produces

$$U_{nj} = \rho(M_n SR_j) + \mu_n SR_j + \theta(PP_j/I_n) + \eta_n PP_j + \varepsilon_{nj}.$$

Since μ_n and η_n are not observed, the terms $\mu_n SR_j$ and $\eta_n PP_j$ become part of the unobserved component of utility,

$$U_{nj} = \rho(M_n SR_j) + \theta(PP_j/I_n) + \tilde{\varepsilon}_{nj},$$

where $\tilde{\varepsilon}_{nj} = \mu_n SR_j + \eta_n PP_j + \varepsilon_{nj}$. The new error terms $\tilde{\varepsilon}_{nj}$ cannot possibly be distributed independently and identically as required for the logit formulation. Since μ_n and η_n enter each alternative, $\tilde{\varepsilon}_{nj}$ is necessarily correlated over alternatives: $Cov(\tilde{\varepsilon}_{nj}, \tilde{\varepsilon}_{nk}) = Var(\mu_n)SR_j SR_k + Var(\eta_n)PP_j PP_k \neq 0$ for any two cars j and k . Furthermore, since SR_j and PP_j vary over alternatives, the variance of $\tilde{\varepsilon}_{nj}$ varies over alternatives, violating the assumption of identically distributed errors: $Var(\tilde{\varepsilon}_{nj}) = Var(\mu_n)SR_j^2 + Var(\eta_n)PP_j^2 + Var(\varepsilon_{nj})$, which is different for different j .

This example illustrates the general point that when tastes vary systematically in the population in relation to observed variables, the variation can be incorporated into logit models. However, if taste

variation is at least partly random, logit is a misspecification. As an approximation, logit might be able to capture the average tastes fairly well even when tastes are random, since the logit formula seems to be fairly robust to misspecifications. The researcher might therefore choose to use logit even when she knows that tastes have a random component, for the sake of simplicity. However, there is no guarantee that a logit model will approximate the average tastes. And, even if it does, logit does not provide information on the distribution of tastes around the average. This distribution can be important in many situations, such as forecasting the penetration of a new product that appeals to a minority of people rather than to the average tastes. To incorporate random taste variation appropriately and fully, a probit or mixed logit model can be used instead.

3.3.2 Substitution patterns

When the attributes of one alternative improve (e.g., its price drops), the probability of its being chosen rises. Some of the people who would have chosen other alternatives under the original attributes now choose this alternative instead. Since probabilities sum to one over alternatives, an increase in the probability of one alternative necessarily means a decrease in probability for other alternatives. The pattern of substitution among alternatives has important implications in many situations. For example, when a cell phone manufacturer launches a new product with extra features, the firm is vitally interested in knowing the extent to which the new product will draw customers away from its other cell phones rather than from competitors' phones, since the firm makes more profit from the later than the former. Also, as we will see, the pattern of substitution affects the demand for a product and the change in demand when attributes change. Substitution patterns are therefore important even when the researcher is only interested in market share without being concerned about where the share comes from.

The logit model implies a certain pattern of substitution across alternatives. If substitution actually occurs in this way given the researcher's specification of representative utility, then the logit model is appropriate. However, to allow for more general patterns of substitution and to investigate which pattern is most accurate, more flexible models are needed. The issue can be seen in either of two ways, as a

restriction on the ratios of probabilities and/or as a restriction on the cross-elasticities of probabilities. We present each way of characterizing the issue below.

The independence from irrelevant alternatives (IIA) property

For any two alternatives i and k , the ratio of the logit probabilities is

$$\begin{aligned}\frac{P_{ni}}{P_{nk}} &= \frac{e^{V_{ni}} / \sum_j e^{V_{nj}}}{e^{V_{nk}} / \sum_j e^{V_{nj}}} \\ &= \frac{e^{V_{ni}}}{e^{V_{nk}}} = e^{V_{ni} - V_{nk}}.\end{aligned}$$

This ratio does not depend on any alternatives other than i and k . That is, the relative odds of choosing i over k is the same no matter what other alternatives are available or what the attributes of the other alternatives are. Since the ratio is independent from alternatives other than i and k , it is said to be independent from “irrelevant” alternatives. The logit model exhibits this “independence from irrelevant alternatives,” or IIA, property.

In many settings, choice probabilities that exhibit IIA provide an accurate representation of reality. In fact, Luce (1959) considered IIA to be a property of appropriately specified choice probabilities. He derived the logit model directly from an assumption that choice probabilities exhibit IIA, rather than (as we have done) derive the logit formula from an assumption about the distribution of unobserved utility and then observe that IIA is a resulting property.

While the IIA property is realistic in some choice situations, it is clearly inappropriate in others, as first pointed out by Chipman (1960) and Debreu (1960). Consider the famous red bus/blue bus problem. A traveler has a choice of going to work by car or taking a blue bus. For simplicity assume that the representative utility of the two modes are the same, such that the choice probabilities are equal: $P_c = P_{bb} = 1/2$, where c is car and bb is blue bus. In this case, the ratio of probabilities is one: $P_c/P_{bb} = 1$.

Now suppose that a red bus is introduced and that the traveler considers the red bus to be exactly like the blue bus. The probability that the traveler will take the red bus is therefore the same as for the blue bus, such that the ratio of their probabilities is one: $P_{rb}/P_{bb} = 1$. However, in the logit model the ratio P_c/P_{bb} is the same whether or

not another alternative, in this case the red bus, exists. This ratio therefore remains at one. The only probabilities for which $P_c/P_{bb} = 1$ and $P_{rb}/P_{bb} = 1$ are $P_c = P_{bb} = P_{rb} = 1/3$, which are the probabilities that the logit model predicts.

In real life, however, we would expect the probability of taking a car to remain the same when a new bus is introduced that is exactly the same as the old bus. We would also expect the original probability of taking bus to be split between the two buses after the second one is introduced. That is, we would expect $P_c = 1/2$ and $P_{bb} = P_{rb} = 1/4$. In this case, the logit model, because of its IIA property, overestimates the probability of taking either of the buses and underestimates the probability of taking a car. The ratio of probabilities of car and blue bus, P_c/P_{bb} , actually changes with the introduction of the red bus, rather than remaining constant as required by the logit model.

This red bus/blue bus example is rather stark and unlikely to be encountered in the real world. However, the same kind of misprediction arises with logit models whenever the ratio of probabilities for two alternatives changes with the introduction or change in another alternative. For example, suppose a new transit mode is added that is similar, but not exactly like, the existing modes, such as an express bus along a line that already has standard bus service. This new mode might be expected to reduce the probability of regular bus by a greater proportion than it reduces the probability of car, such that ratio of probabilities for car and regular bus does not remain constant. The logit model would overpredict demand for the two bus modes in this situation. Other examples are given by, for example, Ortuzar (1983) and Brownstone and Train (1999).

Proportional substitution

The same issue can be expressed in terms of the cross-elasticities of logit probabilities. Let us consider changing an attribute of alternative j . We want to know the effect of this change on the probabilities for all the *other* alternatives. Section (3.6) derives the formula for the elasticity of P_{ni} with respect to a variable that enters the representative utility of alternative j :

$$E_{iz_{nj}} = -\beta_z z_{nj} P_{nj}$$

where z_{nj} is the attribute of alternative j as faced by person n and β_z is its coefficient (or, if the variable enters representative utility non-

linearly, then β_z is the derivative of V_{nj} with respect to z_{nj}).

This cross-elasticity is the same for all i : i does not enter the formula. An improvement in the attributes of an alternative reduces the probabilities for all the other alternatives by the same percent. If one alternative's probability drops by ten percent, then all the other alternatives' probabilities also drop by ten percent (except of course the alternative whose attribute changed; its probability rises due to the improvement.) A way of stating this phenomenon succinctly is that an improvement in one alternative draws proportionately from the other alternatives. Similarly for a decrease in the representative utility of an alternative: the probabilities for all other alternatives rise by the same percent.

This pattern of substitution, which can be called "proportionate shifting," is a manifestation of the IIA property. The ratio of probabilities for alternatives i and k stays constant when an attribute of alternative j changes only if the two probabilities change by the same proportion. With superscript 0 denoting probabilities before the change and 1 after, the IIA property requires that

$$\frac{P_{ni}^1}{P_{nk}^1} = \frac{P_{ni}^0}{P_{nk}^0}$$

when an attribute of alternative j changes. This equality can only be maintained if each probability changes by the same proportion: $P_{ni}^1 = \lambda P_{ni}^0$ and $P_{nk}^1 = \lambda P_{nk}^0$ where both λ 's are the same.

Proportionate substitution can be realistic for some situations, in which case the logit model is appropriate. In many settings, however, other patterns of substitution can be expected, and imposing proportionate substitution through the logit model can lead to unrealistic forecasts. Consider a situation that is important to the California Energy Commission (CEC), which has the responsibility of investigating policies to promote energy efficient vehicles in California and reducing the state's reliance on gasoline for cars. Suppose for the sake of illustration that there are three kinds of vehicles: large gas cars, small gas cars, and small electric cars. Suppose also that under current conditions the probabilities that a household will choose each of these vehicles are .66, .33, and .01, respectively. The CEC is interested in knowing the impact of subsidizing the electric cars. Suppose the subsidy is sufficient to raise the probability for the electric car from .01 to .10. By the logit model, the probability for each of the gas cars would

be predicted to drop by the same percent. The large gas car would drop by ten percent from .66 to .60 and the probability for the small gas car would drop by the same ten percent from .33 to .30. In terms of absolute numbers, the increased probability for the small electric car (.09) is predicted by the logit model to come twice as much from large gas cars (.06) as from small gas cars (0.03).

This pattern of substitution is clearly unrealistic. Since the electric car is small, subsidizing it can be expected to draw more from small gas cars than from large gas cars. In terms of cross-elasticities, we would expect the cross-elasticity for small gas cars with respect to an improvement in small electric cars to be higher than that for large gas cars. This difference is important in the CEC's policy analysis. The logit model will over-predict the gas savings that result from the subsidy, since it over-predicts the substitution away from large gas cars (the "gas guzzlers") and under-predicts the substitution away from small "gas-sipper" cars. From a policy perspective, this misprediction can be critical, causing a subsidy program to seem more beneficial than it actually is. This is the reason that the CEC uses models that are more general than logit to represent substitution across vehicles. The nested logit, probit and mixed logit models of Chapters 4-6 provide viable options for the researcher.

Advantages of IIA

As just discussed, the IIA property of logit can be unrealistic in many settings. However, when the IIA property reflects reality (or an adequate approximation to reality), considerable advantages are gained by its employment. First, because of the IIA property, it is possible to estimate model parameters consistently on a subset of alternatives for each sampled decision-maker. For example, in a situation with 100 alternatives, the researcher might, so as to reduce computer time, estimate on a subset of 10 alternatives for each sampled person, with the person's chosen alternative included as well as 9 alternatives randomly selected from the remaining 99. Since relative probabilities within a subset of alternatives are unaffected by the attributes or existence of alternatives not in the subset, exclusion of alternatives in estimation does not affect the consistency of the estimator. Details of this type of estimation are given in section (3.7.1). This fact has considerable practical importance. In analyzing choice situations for which the number

of alternatives is large, estimation on a subset of alternatives can save substantial amounts of computer time. At an extreme, the number of alternatives might be so large as to preclude estimation altogether if it were not possible to utilize a subset of alternatives.

Another practical use of the IIA property arises when the researcher is only interested in examining choices among a subset of alternatives and not among all alternatives. For example, consider a researcher who is interested in understanding the factors that affect workers' choice between car and bus modes for travel to work. The full set of alternative modes includes walking, bicycling, motor-biking, skate-boarding, etc. If the researcher believed that the IIA property holds adequately well in this case, she could estimate a model with only car and bus as the alternatives and exclude from the analysis sampled workers who used other modes. This strategy would save the researcher considerable time and expense developing data on the other modes without hampering her ability to examine the factors related to car and bus.

Tests of IIA

Whether IIA holds in a particular setting is an empirical question, amenable to statistical investigation. Tests of IIA were first developed by McFadden, Train and Tye (1978). Two types of tests are suggested. First, the model can be re-estimated on a subset of the alternatives. Under IIA, the ratio of probabilities for any two alternatives is the same whether or not other alternatives are available. As a result, if IIA holds in reality, then the parameter estimates obtained on the subset of alternatives will not be significantly different from those obtained on the full set of alternatives. A test of the hypothesis that the parameters on the subset are the same as the parameters on the full set constitutes a test of IIA. Hausman and McFadden (1984) provide an appropriate statistic for this type of test. Second, the model can be re-estimated with new, cross-alternative variables, that is, with variables from one alternative entering the utility of another alternative. If the ratio of probabilities for alternatives i and k actually depends on the attributes and existence of a third alternative j (in violation of IIA), then the attributes of alternative j will enter significantly the utility of alternatives i or k within a logit specification. A test of whether cross-alternative variables enter the model therefore constitutes a test of IIA. McFadden (1987) developed a procedure for performing this kind

of test with regressions: with the dependent variable being the residuals of the original logit model and the explanatory variables being appropriately specified cross-alternative variables. Train, Ben-Akiva and Atherton (1989) show how this procedure can be performed conveniently within the logit model itself.

The advent of models that do not exhibit IIA, and especially the development of software for estimating these models, makes testing IIA easier than before. For more flexible specifications, such as GEV and mixed logit, the simple logit model with IIA is a special case that arises under certain constraints on the parameters of the more flexible model. In these cases, IIA can be tested by testing these constraints. For example, a mixed logit model becomes a simple logit if the mixing distribution has zero variance. IIA can be tested by estimating a mixed logit and testing whether the variance of the mixing distribution is in fact zero.

A test of IIA as a constraint on a more general model necessarily operates under the maintained assumption that the more general model is itself an appropriate specification. The tests on subsets of alternatives (Hausman and McFadden, 1984) and cross-alternative variables (McFadden, 1987; Train et al., 1989), while more difficult to perform, operate under less restrictive maintained hypotheses. The counterpoint to this advantage, of course, is that, when IIA fails, these tests do not provide as much guidance on the correct specification to use instead of logit.

3.3.3 Panel data

In many settings, the researcher can observe numerous choices made by each decision-maker. For example, in labor studies, sampled people are observed to work or not work in each month over several years. Data on the current and past vehicle purchases of sampled households might be obtained by a researcher who is interested in the dynamics of car choice. In market research surveys, respondents are often asked a series of hypothetical choice questions, called “stated preference” experiments. For each experiment, a set of alternative products with different attributes are described and the respondent is asked to state which product he would choose. A series of such questions is asked, with the attributes of the products varying so as to determine how the respondent’s choice changes when the attributes change. The re-

searcher therefore observes the sequence of choices by each respondent. Data that represent repeated choices like these are called panel data.

If the unobserved factors that affect decision-makers are independent over the repeated choices, then logit can be used to examine panel data in the same way as purely cross-sectional data. Any dynamics related to observed factors that enter the decision-process, such as state-dependence (by which the person's past choices influence their current choices) or lagged response to changes in attributes, can be accommodated. However, dynamics associated with unobserved factors cannot be handled, since the unobserved factors are assumed to be unrelated over choices.

The utility that decision-maker n obtains from alternative j in period or choice situation t is

$$U_{njt} = V_{njt} + \varepsilon_{njt} \quad \forall j, t.$$

If ε_{njt} is distributed extreme value, independent over n , j , and, importantly, t , then, using the same proof as for (3.6), the choice probabilities are:

$$P_{nit} = \frac{e^{V_{nit}}}{\sum_j e^{V_{njt}}}. \quad (3.9)$$

Each choice situation by each decision-maker becomes a separate observation. If representative utility for each period is specified to depend only on variables for that period, e.g., $V_{njt} = \beta' x_{njt}$ where x_{njt} is a vector of variables describing alternative j as faced by n in period t , then there is essentially no difference in the logit model with panel data than with purely cross-sectional data.

Dynamic aspects of behavior can be captured by specifying representative utility in each period to depend on observed variables from other periods. For example, a lagged price response is represented by entering the price in period $t - 1$ as an explanatory variable in the utility for period t . Prices in future periods can be entered, as by Adamowicz (1994), to capture consumers' anticipation of future price changes. Under the assumptions of the logit model, the dependent variable in previous periods can also be entered as an explanatory variable. Suppose for example that there is inertia, or habit formation, in people's choices such that they tend to stay with the alternative that they have previously chosen unless another alternative provides sufficiently higher utility to warrant a switch. This behavior is captured as $V_{njt} = \alpha y_{nj(t-1)} + \beta x_{njt}$ where $y_{njt} = 1$ if n chose j in period t and

zero otherwise. With $\alpha > 0$, the utility of alternative j in the current period is higher if alternative j was consumed in the previous period. The same specification can also capture a type of variety seeking. If α is negative, the consumer obtains higher utility from *not* choosing the same alternative that he chose in the last period. Numerous variations on these concepts are possible. Adamowicz (1994) enters the *number* of times the alternative has been chosen previously, rather than simply a dummy for the immediately previous choice. Erdem (1996) enters the *attributes* of previously chosen alternatives, with the utility of each alternative in the current period depending on the similarity of its attributes to these previously experienced attributes.

The inclusion of the lagged dependent variable does not induce inconsistency in estimation since, for a logit model, the errors are assumed to be independent over time. The lagged dependent variable $y_{nj(t-1)}$ is uncorrelated with the current error ε_{njt} due to this independence. The situation is analogous to linear regression models, where a lagged dependent variable can be added without inducing bias as long as the errors are independent over time.

Of course, the assumption of independent errors over time is severe. Usually, one would expect that there are some factors that are not observed by the researcher that affect each of the decision-makers' choices. In particular, if there are dynamics in the observed factors, then the researcher might expect there to be dynamics in the unobserved factors as well. In these situations, the researcher can either use a model such as probit or mixed logit that allows unobserved factors to be correlated over time, or re-specify representative utility to bring the sources of the unobserved dynamics into the model explicitly such that the remaining errors are independent over time.

3.4 Non-linear representative utility

In some contexts, the researcher will find it useful to allow parameters to enter representative utility nonlinearly. Estimation is more difficult since the log-likelihood function might not be globally concave, and computer routines are not as widely available as for logit models with linear-in-parameters utility. However, the aspects of behavior that the researcher is investigating might include parameters that are interpretable only when they enter utility non-linearly. In these cases, the effort of writing one's own code can be warranted. Two examples

illustrate this point.

Example 1: the goods/leisure trade-off

Consider a workers' choice of mode (car or bus) for trips to work. Suppose that workers also choose the number of hours to work based on the standard trade-off between goods and leisure. Train and McFadden (1978) developed a procedure for examining these interrelated choices. As we see below, the parameters of the workers' utility function over goods and leisure enter non-linearly in the utility for modes of travel.

Assume that workers' preferences regarding goods G and leisure L are represented by a Cobb-Douglas utility function of the form:

$$U = (1 - \beta)\ln G + \beta\ln L.$$

The parameter β reflects the worker's relative preference for goods and leisure, with higher β implying greater preference for leisure relative to goods. Each worker has a fixed amount of time (24 hours a day) and faces a fixed wage rate, w . In the standard goods-leisure model, the worker chooses the number of hours to work that maximizes U subject to the constraints that (1) the number of hours worked plus the number of leisure hours equals the number of hours available, and (2) the value of goods consumed equals the wage rate times the number of hours worked.

When mode choice is added to the model, the constraints on time and money change. Each mode takes a certain amount of time and costs a certain amount of money. Conditional on choosing car, the worker maximizes U subject to the constrain that (1) the number of hours worked plus the number of leisure hours equals the number of hours available *after the time spent driving to work in the car is subtracted* and (2) the value of goods consumed equals the wage rate times the number of hours worked *minus the cost of driving to work*. The utility associated with choosing to travel by car is the highest value of U that can be attained under these constraints. Similarly, the utility of taking the bus to work is the maximum value of U that can be obtained given the time and money that is left after the bus time and cost are subtracted. Train and McFadden derived the maximizing values of U conditional on each mode. For the U given above, these values are

$$U_j = -\alpha((c_j/w^\beta) + w^{1-\beta}t_j) \text{ for } j = \text{car and bus}$$

The cost of travel is divided by w^β and travel time is multiplied by $w^{1-\beta}$. The parameter β , which denotes workers' relative preference for goods and leisure, enters the mode choice utility non-linearly. Since this parameter has meaning, the researcher might want to estimate it within this non-linear utility rather than use a linear-in-parameters approximation.

Example 2: geographic aggregation

Models have been developed and widely used for travelers' choice of destination for various types of trips, such as shopping trips, within a metropolitan area. Usually, the metropolitan area is partitioned into "zones," and the models give the probability that a person will choose to travel to a particular zone. Representative utility for each zone depends on the time and cost of travel to the zone plus a variety of variables, such as residential population and retail employment, that reflect reasons that people might want to visit the zone. These latter variables are called "attraction" variables; label them by the vector a_j for zone j . Since it is these attraction variables that give rise to parameters entering non-linearity, assume for simplicity that representative utility depends only on these variables.

The difficulty in specifying representative utility comes in recognizing that the researcher's decision of how large an area to include in each zone is fairly arbitrary. It would be useful to have a model that is not sensitive to the level of aggregation in the zonal definitions. If two zones are combined, it would be useful for the model to give a probability of traveling to the combined zone that is the same as the sum of the probabilities of traveling to the two original zones. This consideration places restrictions on the form of representative utility.

Consider zones j and k which, when combined, are labeled zone c . Population and employment in the combined zone are necessarily the sum of that in the two original zones: $a_j + a_k = a_c$. In order for the models to give the same probability for choosing these zones before and after their merger, the model must satisfy:

$$P_{nj} + P_{nk} = P_{nc}$$

which for logit models takes the form

$$\frac{e^{V_{nj}} + e^{V_{nk}}}{e^{V_{nj}} + e^{V_{nk}} + \sum_{\ell \neq j,k} e^{V_{n\ell}}} = \frac{e^{V_{nc}}}{e^{V_{nc}} + \sum_{\ell \neq j,k} e^{V_{n\ell}}}$$

This equality holds only when $\exp(V_{nj}) + \exp(V_{nk}) = \exp(V_{nc})$. If representative utility is specified as $V_{n\ell} = \ln(\beta' a_\ell)$ for all zones ℓ , then the inequality holds: $\exp(\ln(\beta' a_j)) + \exp(\ln(\beta' a_k)) = \beta' a_j + \beta' a_k = \beta' a_c = \exp(\ln(\beta' a_c))$. Therefore, to specify a destination choice model that is not sensitive to the level of zonal aggregation, representative utility needs to be specified with parameters inside a log operation.

3.5 Consumer Surplus

For policy analysis, the researcher is often interested in measuring the change in consumer surplus that is associated with a particular policy. For example, if a new alternative is being considered, such as building a light rail system in a city, then it is important to measure the benefits of the project to see if they warrant the costs. Similarly, a change in the attributes of an alternative can have an impact on consumer surplus that is important to assess. Degradation of the water quality of rivers harms the anglers who can no longer fish as effectively at the damaged sites. Measuring this harm in monetary terms is a central element of legal action against the polluter. Often the distributional impacts of a policy are important to assess, such as how the burden of a tax is borne by different population groups.

Under the logit assumptions, the consumer surplus associated with a set of alternatives takes a closed form that is easy to calculate. By definition, a person's consumer surplus is the utility, in dollar terms, that the person receives in the choice situation. The decision-maker chooses the alternative that provides the greatest utility. Consumer surplus is therefore $CS_n = (1/\alpha_n) \max_j (U_{nj} \forall j)$, where α_n is the marginal utility of income: $dU_n/dY_n = \alpha_n$ with Y_n being the income of person n . The division by α_n translates utility into dollars since $1/\alpha_n = dY_n/dU_n$. The researcher does not observe U_{nj} and therefore cannot use this expression to calculate the decision-maker's consumer surplus. Instead, the researcher observes V_{nj} and knows the distribution of the remaining portion of utility. With this information, the researcher is able to calculate the expected consumer surplus:

$$E(CS_n) = \frac{1}{\alpha_n} E[\max(V_{nj} + \varepsilon_{nj} \forall j)],$$

where the expectation is over all possible values of the ε_{nj} 's. Williams (1977) and Small and Rosen (1981) show that, if each ε_{nj} is iid extreme

value and utility is linear in income (such that α_n is constant with respect to income), then this expectation becomes:

$$E(CS_n) = \frac{1}{\alpha_n} \ln \left(\sum_{j=1}^J e^{V_{nj}} \right) + C. \quad (3.10)$$

where C is an unknown constant that represents the fact that the absolute level of utility cannot be measured. As we see below, this constant is irrelevant from a policy perspective and can be ignored.

Note that the term in parentheses in this expression is the denominator of the logit choice probability (3.6). Aside from the division and addition of constants, expected consumer surplus in a logit model is simply the log of the denominator of the choice probability. It is often called “the log-sum term.” This relation between the two formulas has no economic meaning, in the sense that there is nothing about a denominator in a choice probability that makes it necessarily related to consumer surplus. It is simply the outcome of the mathematical form of the extreme value distribution. However, the relation makes calculation of expected consumer surplus very easy, which is another of the many conveniences of logit.

Under the standard interpretation for the distribution of errors, as described in the last paragraph of 2.3, $E(CS_n)$ is the average consumer surplus in the subpopulation of people who have the same representative utilities as person n . Total consumer surplus in the population is calculated as the weighted sum of $E(CS_n)$ over a sample of decision-makers, with the weights reflecting the number of people in the population who face the same representative utilities as the sampled person.

The change in consumer surplus that results from a change in the alternatives and/or the choice set is calculated from (3.10). In particular, $E(CS_n)$ is calculated twice: first under the conditions before the change and again under the conditions after the change. The difference between the two results is the change in consumer surplus:

$$\Delta E(CS_n) = \frac{1}{\alpha_n} \left[\ln \left(\sum_{j=1}^{J^1} e^{V_{nj}^1} \right) - \ln \left(\sum_{j=1}^{J^0} e^{V_{nj}^0} \right) \right],$$

where the superscripts 0 and 1 refer to before and after the change. The number of alternatives can change (e.g., a new alternative added) as well as the attributes of the alternatives. Since the unknown constant C enters expected consumer surplus both before and after the

change, it drops out of the difference and can therefore be ignored when calculating changes in consumer surplus.

To calculate the change in consumer surplus, the researcher must know or have estimated the marginal utility of income α_n . Usually a price or cost variable enters representative utility, in which case the negative of its coefficient is α_n by definition. (A price or cost coefficient is negative; the negative of a negative coefficient gives a positive α_n). For example, in the choice between car and bus, utility is $U_{nj} = \beta_1 t_{nj} + \beta_2 c_{nj}$ where t is time, c is cost, and both β_1 and β_2 are negative, indicating that utility decreases as the time or cost for a trip increases. The negative of the cost coefficient, $-\beta_2$, is the amount that utility rises due to a one-dollar decrease in costs. A one-dollar reduction in costs is equivalent to a one-dollar increase in income, since the person gets to spend the dollar that he saves in travel costs just the same as if he got the extra dollar in income. The amount $-\beta_2$ is therefore the increase in utility from a one-dollar increase in income: the marginal utility of income. It is the same amount in this case for all n . If c_{nj} entered representative utility interacting with characteristics of the person other than income, such as $c_{nj}H_n$ where H_n is household size, then the marginal utility of income would be $-\beta_2 H_n$, which varies over n .

Throughout this discussion, α_n has been assumed to be fixed for a given person independent of their income. The formula (3.10) for expected consumer surplus depends critically on the assumption that the marginal utility of income is constant with respect to income. If the marginal utility of income changes with income, then a more complicated formula is needed, since α_n itself becomes a function of the change in attributes. McFadden (1999) and Karlstrom (2000) provide procedures for calculating changes in consumer surplus under these conditions.

The conditions for using expression (3.10) are actually less severe than stated. Since only changes in consumer surplus are relevant for policy analysis, formula (3.10) can be used if the marginal utility of income is constant over the range of implicit income changes that are considered by the policy. Thus, for policy changes that change consumer surplus by small amounts per person relative to their income, the formula can be used even though the marginal utility of income in reality varies with income.

The assumption that α_n does not depend on income has implication

for the specification of representative utility. As discussed above, α_n is usually taken as the absolute value of the coefficient of price or cost. Therefore, if the researcher plans to use her model to estimate changes in consumer surplus and wants to apply formula (3.10), this coefficient cannot be specified to depend on income. In the mode choice example, cost can be multiplied by household size such that the cost coefficient, and hence the marginal utility of income, varies over households of different size. However, if cost is divided by the household's income, then the coefficient of cost depends on income, violating the assumption needed for expression (3.10). This violation might not be important for small changes in consumer surplus, but certainly becomes important for large changes.

3.6 Derivatives and Elasticities

Since choice probabilities are a function of observed variables, it is often useful to know the extent to which these probabilities change in response to a change in some observed factor. For example, in a household's choice of make and model of car to buy, a natural question is: to what extent will the probability of choosing a given car increase if the vehicle's fuel efficiency is improved? From competing manufacturers points of view, a related question is: to what extent will the probability of households' choosing, say, a Toyota decrease if the fuel efficiency of a Honda improves.

To address these questions, derivatives of the choice probabilities are calculated. The change in the probability that decision-maker n chooses alternative i given a change in an observed factor, z_{ni} , entering the representative utility of that alternative (and holding the representative utility of other alternatives constant) is

$$\begin{aligned}
 \frac{\partial P_{ni}}{\partial z_{ni}} &= \frac{\partial [(e^{V_{ni}})/(\sum_j e^{V_{nj}})]}{\partial z_{ni}} \\
 &= [e^{V_{ni}} / \sum e^{V_{nj}}] \left(\frac{\partial V_{ni}}{\partial z_{ni}} \right) - [e^{V_{ni}} / (\sum e^{V_{nj}})^2] (e^{V_{ni}}) \left(\frac{\partial V_{ni}}{\partial z_{ni}} \right) \\
 &= \frac{\partial V_{ni}}{\partial z_{ni}} (P_{ni} - P_{ni}^2) \\
 &= \frac{\partial V_{ni}}{\partial z_{ni}} P_{ni} (1 - P_{ni}).
 \end{aligned}$$

If representative utility is linear in z_{ni} with coefficient β_z , the derivative

becomes $\beta_z P_{ni}(1 - P_{ni})$. This derivative is largest when $P_{ni} = 1 - P_{ni}$, which occurs when $P_{ni} = .5$. It becomes smaller as P_{ni} approaches zero or one. The sigmoid probability curve in Figure 3.1 is consistent with these facts. Stated intuitively, the effect of a change in an observed variable is highest when the choice probabilities indicate a high degree of uncertainty regarding the choice. As the choice becomes more certain (i.e., the probabilities approach zero or one), the effect of a change in an observed variable lessens.

One can also determine the extent to which the probability of choosing a particular alternative changes when an observed variable relating to *another* alternative changes. Let z_{nj} denote an attribute of alternative j . How does the probability of choosing alternative i change as z_{nj} increases?

$$\begin{aligned} \frac{\partial P_{ni}}{\partial z_{nj}} &= \frac{\partial[(e^{V_{ni}})/(\sum_k e^{V_{nk}})]}{\partial z_{nj}} \\ &= -[(e^{V_{ni}})/(\sum_k e^{V_{nk}})^2](e^{V_{nj}})(\frac{\partial V_{nj}}{\partial z_{nj}}) \\ &= -\frac{\partial V_{nj}}{\partial z_{nj}} P_{ni} P_{nj}. \end{aligned}$$

When V_{nj} is linear in z_{nj} with coefficient β_z , then this cross-derivative becomes $-\beta_z P_{ni} P_{nj}$. If z_{nj} is a desirable attribute such that β_z is positive, then raising z_{nj} decreases the probability of choosing each alternative other than j . Furthermore, the decrease in probability is proportional to the value of the probability before z_{nj} was changed.

A logically necessary aspect of derivatives of choice probabilities is that, when an observed variable changes, the changes in the choice probabilities sum to zero. This is a consequence of the fact that the probabilities must sum to one before and after the change; it is demonstrated for logit models as follows:

$$\begin{aligned} \sum_{i=1}^J \frac{\partial P_{ni}}{\partial z_{nj}} &= (\frac{\partial V_{nj}}{\partial z_{nj}}) P_{nj} (1 - P_{nj}) + \sum_{i \neq j} (-\frac{\partial V_{nj}}{\partial z_{nj}}) P_{nj} P_{ni} \\ &= (\frac{\partial V_{nj}}{\partial z_{nj}}) P_{nj} \left[(1 - P_{nj}) - \sum_{i \neq j} P_{ni} \right] \\ &= (\frac{\partial V_{nj}}{\partial z_{nj}}) P_{nj} [(1 - P_{nj}) - (1 - P_{nj})] \end{aligned}$$

$$= 0.$$

In practical terms, if one alternative is improved so that the probability of its being chosen increases, the additional probability is necessarily “drawn” from other alternatives. To increase the probability of one alternative necessitates decreasing the probability of another alternative. While obvious, this fact is often forgotten by planners who want to improve demand for one alternative without reducing demand for other alternatives.

Economists often measure response by elasticities rather than derivatives, since elasticities are normalized for the variables’ units. An elasticity is the percent change in one variable that is associated with a percent change in another variable. The elasticity of P_{ni} with respect to z_{ni} , a variable entering the utility of alternative i , is

$$\begin{aligned} E_{iz_{ni}} &= (\partial P_{ni} / \partial z_{ni})(z_{ni} / P_{ni}) \\ &= \frac{\partial V_{ni}}{\partial z_{ni}} P_{ni} (1 - P_{ni}) (z_{ni} / P_{ni}) \\ &= \frac{\partial V_{ni}}{\partial z_{ni}} z_{ni} (1 - P_{ni}). \end{aligned}$$

If representative utility is linear in z_{ni} with coefficient β_z , then $E_{iz_{ni}} = \beta_z z_{ni} (1 - P_{ni})$.

The cross-elasticity of P_{ni} with respect to a variable entering alternative j is

$$\begin{aligned} E_{iz_{nj}} &= (\partial P_{ni} / \partial z_{nj})(z_{nj} / P_{ni}) \\ &= -\frac{\partial V_{nj}}{\partial z_{nj}} z_{nj} P_{nj}. \end{aligned}$$

which in the case of linear utility reduces to $E_{iz_{nj}} = -\beta_z z_{nj} P_{nj}$. As discussed in section (3.3.2), this cross elasticity is the same for all i : a change in an attribute of alternative j changes the probabilities for all other alternatives by the same percent. This property of the logit cross-elasticities is a manifestation, or re-statement, of the IIA property of the logit choice probabilities.

3.7 Estimation

Manski and McFadden (1981) and Cosslett (1981) describe estimation methods under a variety of sampling procedures. We discuss in this section estimation under the most prominent of these sampling schemes. We first describe estimation when the sample is exogenous and all alternatives are used in estimation. We then discuss estimation on a subset of alternatives and with certain types of choice-based (i.e., non-exogenous) samples.

3.7.1 Exogenous sample

Consider first the situation in which the sample is exogenously drawn, that is, is either random or stratified random with the strata defined on factors that are exogenous to the choice being analyzed. If the sampling procedure is related to the choice being analyzed (for example, if mode choice is being examined and the sample is drawn by selecting people on buses and pooling them with people selected at toll booths), then more complex estimation procedures are generally required, as discussed in the next section. We also assume that the explanatory variables are exogenous to the choice situation. That is, the variables entering representative utility are independent of the unobserved component of utility.

A sample of N decision-makers is obtained for the purposes of estimation. Since the logit probabilities take a closed form, the traditional maximum likelihood procedures can be applied. The probability of person n choosing the alternative that he was actually observed to choose can be expressed as

$$\prod_i (P_{ni})^{y_{ni}},$$

where $y_{ni} = 1$ if person n chose i and zero otherwise. Note that since $y_{ni} = 0$ for all nonchosen alternatives and P_{ni} raised to the power of zero is 1, this term is simply the probability of the chosen alternative.

Assuming that each decision-maker's choice is independent of that of other decision-makers, the probability of each person in the sample choosing the alternative that he was observed actually to choose is

$$L(\beta) = \prod_{n=1}^N \prod_i (P_{ni})^{y_{ni}}$$

where β is a vector containing the parameters of the model. The log-likelihood function is then

$$LL(\beta) = \sum_{n=1}^N \sum_i y_{ni} \ln(P_{ni}) \quad (3.11)$$

and the estimator is the value of β that maximizes this function. McFadden (1974) shows that $LL(\beta)$ is globally concave for linear-in-parameters utility, and many statistical packages are available for estimation of these models. When parameters enter representative utility non-linearly, the researcher might need to write her own estimation code using the procedures described in Chapter 8.

Maximum likelihood estimation in this situation can be re-expressed and re-interpreted in a way that assists in understanding the nature of the estimates. At the maximum of the likelihood function, its derivative with respect to each of the parameters is zero:

$$\frac{dLL(\beta)}{d\beta} = 0. \quad (3.12)$$

The maximum likelihood estimates are therefore the values of β that satisfy this first-order condition. For convenience, let representative utility be linear in parameters: $V_{nj} = \beta' x_{nj}$. This specification is not required, but makes the notation and discussion more succinct. Using (3.11) and the formula for the logit probabilities, we show at the end of this subsection that the first order condition (3.12) becomes:

$$\sum_n \sum_i (y_{ni} - P_{ni}) x_{ni} = 0. \quad (3.13)$$

Rearranging and dividing both sides by N , we have

$$\frac{1}{N} \sum_n \sum_i y_{ni} x_{ni} = \frac{1}{N} \sum_n \sum_i P_{ni} x_{ni} \quad (3.14)$$

This expression is readily interpretable. Let $\bar{x}$ denote the average of x over the chosen alternatives by the sampled individuals: $\bar{x} = (1/N) \sum_n \sum_i y_{ni} x_{ni}$. Let $\hat{x}$ be the average of x over the predicted choices of the sampled decision-makers: $\hat{x} = (1/N) \sum_n \sum_i P_{ni} x_{ni}$. The observed average of x in the sample is $\bar{x}$, while $\hat{x}$ is the predicted average. By (3.14), these two averages equal each other at the maximum likelihood estimates. That is: the maximum likelihood estimates of β

are those that make the predicted average of each explanatory variable equal to the observed average in the sample. In this sense, the estimates induce the model to reproduce the observed averages in the sample.

This property of the maximum likelihood estimator for logit models takes a special meaning for the alternative-specific constants. An alternative-specific constant is the coefficient of a dummy variable that identifies an alternative. A dummy for alternative j is a variable whose value in the representative utility of alternative i is $d_i^j = 1$ for $i = j$ and zero otherwise. By (3.14), the estimated constant is the one that gives

$$\begin{aligned} \frac{1}{N} \sum_n \sum_i y_{ni} d_i^j &= \frac{1}{N} \sum_n \sum_i P_{ni} d_i^j \\ S_j &= \hat{S}_j \end{aligned}$$

where S_j is the share of people in the sample who chose alternative j and $\hat{S}_j$ is the predicted share for alternative j . With alternative-specific constants, the predicted shares for the sample equal the observed shares. The estimated model is therefore correct on average within the sample. This feature is similar to the function of a constant in a linear regression model, where the constant assures that the average of the predicted value of the dependent variable equals its observed average in the sample.

The first-order condition (3.13) provides yet another important interpretation. The difference between a person's actual choice, y_{ni} , and the probability of that choice, P_{ni} , is a modeling error, or residual. The left hand side of (3.13) is the sample covariance of the residuals with the explanatory variables. The maximum likelihood estimates are therefore the values of the β 's that make this covariance zero, that is, make the residuals uncorrelated with the explanatory variables. This condition for logit estimates is the same as applies in linear regression models. For a regression model $y_n = \beta' x_n + \varepsilon_n$, the ordinary least squares estimates are the values of β that set $\sum_n (y_n - \beta' x_n) x_n = 0$. This fact is verified by solving for β : $\beta = (\sum_n x_n x_n')^{-1} (\sum_n x_n y_n)$, which is the formula for the ordinary least squares estimator. Since $y_n - \beta' x_n$ is the residual in the regression model, the estimates make the residuals uncorrelated with the explanatory variables.

Under this interpretation, the estimates can be motivated as providing a sample-analog to population characteristics. We have assumed

that the explanatory variables are exogenous, meaning that they are uncorrelated in the population with the model errors. Since the variables and errors are uncorrelated in the population, it makes sense to choose estimates that make the variables and residuals uncorrelated in the sample. The estimates do exactly that: they provide a model that reproduces in the sample the zero covariances that occur in the population.

Estimators that solve equations of the form (3.13) are called method-of-moments estimators, since they use moment conditions (correlations in this case) between residuals and variables to define the estimator. We will return to these estimators when discussing simulation-assisted estimation in Chapter 10.

We asserted without proof that (3.13) is the first-order condition for the maximum likelihood estimator of the logit model. We give that proof now. The log-likelihood function (3.11) can be re-expressed as

$$\begin{aligned}
 LL(\beta) &= \sum_n \sum_i y_{ni} \ln(P_{ni}) \\
 &= \sum_n \sum_i y_{ni} \ln\left(\frac{e^{\beta' x_{ni}}}{\sum_j e^{\beta' x_{nj}}}\right) \\
 &= \sum_n \sum_i y_{ni}(\beta' x_{ni}) - \sum_n \sum_i y_{ni} \ln\left(\sum_j e^{\beta' x_{nj}}\right).
 \end{aligned}$$

The derivative of the log-likelihood function then becomes

$$\begin{aligned}
 \frac{dLL(\beta)}{d\beta} &= \frac{\sum_n \sum_i y_{ni}(\beta' x_{ni})}{d\beta} - \frac{\sum_n \sum_i y_{ni} \ln(\sum_j e^{\beta' x_{nj}})}{d\beta} \\
 &= \sum_n \sum_i y_{ni} x_{ni} - \sum_n \sum_i y_{ni} \sum_j P_{nj} x_{nj} \\
 &= \sum_n \sum_i y_{ni} x_{ni} - \sum_n \left(\sum_j P_{nj} x_{nj}\right) \sum_i y_{ni} \\
 &= \sum_n \sum_i y_{ni} x_{ni} - \sum_n \left(\sum_j P_{nj} x_{nj}\right) \\
 &= \sum_n \sum_i (y_{ni} - P_{ni}) x_{ni}
 \end{aligned}$$

Setting this derivative to zero gives the first-order condition (3.13).

Estimation on a subset of alternatives

In some situations, the number of alternatives facing the decision-maker is so large that estimating model parameters is very expensive or even impossible. With a logit model, estimation can be performed on a subset of alternatives without inducing inconsistency. For example, a researcher examining a choice situation that involves 100 alternatives can estimate on a subset of 10 alternatives for each sampled decision-maker, with the person's chosen alternative included as well as 9 alternatives randomly selected from the remaining 99. If all alternatives have the same chance of being selected into the subset, then estimation proceeds on the subset of alternatives as if it were the full set. If alternatives have unequal probability of being selected, more complicated estimation procedures may be required. The procedure is described as follows.

Suppose that the researcher has used some specific method for randomly selecting alternatives into the subset that is used in estimation for each sampled decision-maker. Denote the full set of alternatives as F and a subset of alternatives as K . Let $q(K | i)$ be the probability under the researcher's selection method that subset K is selected given that the decision-maker chose alternative i . Assuming that the subset necessarily includes the chosen alternative, $q(K | i) = 0$ for any K that does not include i . The probability that person n chooses alternative i from the full set is P_{ni} . Our goal is to derive a formula for the probability that the person chooses alternative i *conditional* on the researcher selecting subset K for him. This conditional probability is denoted $P_n(i | K)$.

This conditional probability is derived as follows. The joint probability that the researcher selects subset K and the decision-maker chooses alternative i is $Prob(K, i) = q(K | i)P_{ni}$. The joint probability can also be expressed with the opposite conditioning as $Prob(K, i) = P_n(i | K)Q(K)$ where $Q(K) = \sum_{j \in F} P_{nj}q(K | j)$ is the probability of the researcher selecting subset K marginal over all the alternatives that the person could choose. Equating these two expressions and solving for $P_n(i | K)$, we have

$$\begin{aligned} P_n(i | K) &= \frac{P_{ni}q(K | i)}{\sum_{j \in F} P_{nj}q(K | j)} \\ &= \frac{e^{V_{ni}}q(K | i)}{\sum_{j \in F} e^{V_{nj}}q(K | j)} \end{aligned}$$

$$= \frac{e^{V_{ni}} q(K | i)}{\sum_{k \in K} e^{V_{nk}} q(K | j)} \quad (3.15)$$

where the second line has canceled out the denominators of P_{ni} and $P_{nj} \forall j$, and the third equality uses the fact that $q(K | j) = 0$ for any j not in K .

Suppose that the researcher has designed the selection procedure such that $q(K | j)$ is the same for all $j \in K$. This property occurs if, for example, the researcher assigns an equal probability of selection to all non-chosen alternatives, such that the probability of selecting j into the subset when i is chosen by the decision-maker is the same as for selecting i into the subset when j is chosen. McFadden (1978) calls this the “uniform conditioning property” since the subset of alternatives has a uniform (equal) probability of being selected conditional on any of its members being chosen by the decision-maker. When this property is satisfied, $q(K | j)$ cancels out of the above expression, and the probability becomes

$$P_n(i | K) = \frac{e^{V_{ni}}}{\sum_{j \in K} e^{V_{nj}}}$$

which is simply the logit formula for a person who faces the alternatives in subset K .

The conditional likelihood function under the uniform conditioning property is

$$CLL(\beta) = \sum_n \sum_{i \in K_n} y_{ni} \ln \frac{e^{V_{ni}}}{\sum_{j \in K_n} e^{V_{nj}}}$$

where K_n is the subset selected for person n . This function is the same as the log-likelihood function given in (3.11) except that the subset of alternatives K_n replaces, for each sampled person, the complete set. Maximization of CLL provides a consistent estimator of β . However, since information is excluded from CLL that LL incorporates (i.e., information on alternatives not in each subset), the estimator based on CLL is not efficient.

Suppose that the researcher designs a selection process that does not exhibit the uniform conditioning property. In this case, the probability $q(K | i)$ can be incorporated into the model as a separate variable. The expression in (3.15) can be re-written as

$$P_n(i | K) = \frac{e^{V_{ni} + \ln(q(K|i))}}{\sum_{j \in K} e^{V_{nj} + \ln(q(K|j))}}.$$

A variable z_{nj} calculated as $\ln(q(K_n | j))$ is added to the representative utility of each alternative. The coefficient of this variable is constrained to 1 in estimation.

The question arises, why would a researcher ever want to design a selection procedure that does not satisfy the uniform conditioning property, since satisfying the property makes estimation so straightforward? An illustration of the potential benefit of non-uniform conditioning is provided by Train, McFadden and Ben-Akiva (1987) in their study of telecommunications demand. The choice situation in their application included an enormous number of alternatives representing portfolios of calls by time of day, distance and duration. The vast majority of alternatives were hardly ever chosen by anyone in the population. If alternatives had been selected with equal probability for each alternative, it was quite likely that the resulting subsets would consist nearly entirely of alternatives that were hardly ever chosen, coupled with the person's chosen alternative. Comparing a person's chosen alternative to a group of highly undesirable alternatives provides little information about the reasons for a person's choice. To avoid this potential problem, alternatives were selected in proportion to the shares for the alternatives in the population (or, to be precise, estimates of the population shares). This procedure increased the chance that relatively desirable alternatives would be in each subset of alternatives that was used in estimation.

3.7.2 Choice-based samples

In some situations, a sample drawn on the basis of exogenous factors would include few people who have chosen particular alternatives. For example, in the choice of water heaters, a random sample of households in most areas would include only a small number who had chosen solar water heating systems. If the researcher is particularly interested in factors that affect the penetration of solar devices, a random sample would need to be very large to assure a reasonable number of households with solar heat.

In situations such as these, the researcher might instead select the sample, or part of the sample, on the basis of the choice being analyzed. For example, the researcher examining water heaters might supplement a random sample of households with households that are known (perhaps through sales records at stores if the researcher has

access to these records) to have recently installed solar water heaters.

Samples selected on the basis of decision-makers' choices can be purely choice-based or a hybrid of choice-based and exogenous. In a purely choice-based sample, the population is divided into those that choose each alternative and decision-makers are drawn randomly within each group, though at different rates. For example, a researcher who is examining the choice of home location and is interested in identifying the factors that contribute to people choosing one particular community might draw randomly from within that community at the rate of one out of L households, and draw randomly from all other communities at a rate of one out of M , where M is larger than L . This procedure assures that the researcher has an adequate number of people in the sample from the area of interest. A hybrid sample is like the one drawn by the researcher interested in solar water heating, in which an exogenous sample is supplemented with a sample drawn on the basis of the households' choices.

Estimation of model parameters with samples drawn at least partially on the basis of the decision-maker's choice is fairly complex in general, and varies with the exact form of the sampling procedure. For interested readers, Ben-Akiva and Lerman (1985, pp.234-244) provide a useful discussion. One result is particularly significant, since it allows researchers to estimate logit models on choice-based samples without becoming involved in complex estimation procedures. This result, due to Manski and Lerman (1977), can be stated as follows. If the researcher is using a *purely* choice-based sample and includes an alternative-specific constant in the representative utility for each alternative, then estimating a logit model as if the sample were exogenous produces consistent estimates for all the model parameters except the alternative-specific constants. Furthermore, these constants are biased by a known factor and can therefore be adjusted so that the adjusted constants are consistent. In particular, the expectation of the estimated constant for alternative j , labeled $\hat{\alpha}_j$, is related to the true constant α_j^* ,

$$E(\hat{\alpha}_j) = \alpha_j^* - \ln(A_j/S_j)$$

where A_j is the share of decision-makers in the population who chose alternative j and S_j is the share in the choice-based sample who chose alternative j . Consequently, if A_j is known (that is, if population shares are known for each alternative), then a consistent estimate of the alternative-specific constant is the constant $\hat{\alpha}_j$ that is estimated

on the choice-based sample *plus* the log of the ratio of the population share to the sample share.

3.8 Goodness of Fit and Hypothesis Testing

We discuss goodness of fit and hypothesis testing in the context of logit models, where the log likelihood function is calculated exactly. The concepts apply to other models, with appropriate adjustment for simulation variance, when the log likelihood function is simulated rather than calculated exactly.

3.8.1 Goodness of fit

A statistic called the “likelihood ratio index” is often used with discrete choice models to measure how well the models fit the data. Stated more precisely, the statistic measures how well the model, with its estimated parameters, performs compared with a model in which all the parameters are zero (which is usually equivalent to having no model at all). This comparison is made on the basis of the log likelihood function, evaluated at both the estimated parameters and at zero for all parameters.

The likelihood ratio index is defined as

$$\rho = 1 - \frac{LL(\hat{\beta})}{LL(0)},$$

where $LL(\hat{\beta})$ is the value of the log likelihood function at the estimated parameters and $LL(0)$ is its value when all the parameters are set equal to zero. If the estimated parameters do no better, in terms of the likelihood function, than zero parameters (that is, if the estimated model is no better than no model), then $LL(\hat{\beta}) = LL(0)$ and so $\rho = 0$. This is the lowest value that ρ can take (since if $LL(\hat{\beta})$ is less than $LL(0)$, then $\hat{\beta}$ would not be the maximum likelihood estimate).

At the other extreme, suppose the estimated model were so good that each sampled decision-maker’s choice could be predicted perfectly. In this case, the likelihood function at the estimated parameters would be one, since the probability of observing the choices that were actually made is one. And, since the log of one is zero, the log likelihood function would be zero at the estimated parameters. With $LL(\hat{\beta}) = 0$,

$\rho = 1$. This is the highest value that ρ can take. In summary, the likelihood ratio index ranges from zero, when the estimated parameters are no better than zero parameters, to one, when the estimated parameters perfectly predict the choices of the sampled decision-makers.

It is important to note that the likelihood ratio index is not at all similar in its interpretation to the R-squared used in regression, despite both statistics having the same range. R-squared indicates the percent of the variation in the dependent variable that is "explained" by the estimated model. The likelihood ratio has no intuitively interpretable meaning for values between the extremes of zero and one. It is the percent increase in the log likelihood function above the value taken at zero parameters (since $\rho = 1 - (LL(\hat{\beta})/LL(0)) = (LL(0) - LL(\hat{\beta}))/LL(0)$.) However, the meaning of such a percent increase is not clear. In comparing two models estimated on the same data and with the same set of alternatives (such that $LL(0)$ is the same for both models), it is usually valid to say that the model with the higher ρ fits the data better. But this is saying no more than that increasing the value of the log likelihood function is preferable. Two models estimated on samples that are not identical or with a different set of alternatives for any sampled decision maker cannot be compared via their likelihood ratio index values.

Another goodness-of-fit statistic that is sometimes used, but should actually be avoided, is the "percent correctly predicted." This statistic is calculated by identifying for each sampled decision-maker the alternative with the highest probability, based on the estimated model, and determining whether or not this was the alternative that the decision-maker actually chose. The percent of sampled decision-makers for which the highest probability alternative and the chosen alternative are the same is called the percent correctly predicted.

This statistic incorporates a notion that is opposed to the meaning of probabilities and the purpose of specifying choice probabilities. The statistic is based on the idea that the decision-maker is predicted by the researcher to choose the alternative for which the model gives the highest probability. However, as discussed in the derivation of choice probabilities in Chapter 2, the researcher does not have enough information to predict the decision-maker's choice. The researcher has only enough information to state the probability that the decision-maker will choose each alternative. In stating choice probabilities, the researcher is saying that if the choice situation were repeated numerous

times (or faced by numerous people with the same attributes), each alternative would be chosen a certain proportion of the time. This is quite different from saying that the alternative with the highest probability will be chosen each time.

An example might be useful. Suppose an estimated model predicts choice probabilities of .75 and .25 in a two-alternative situation. Those probabilities mean that if 100 people faced the representative utilities that gave these probabilities (or one person faced these representative utilities 100 times), the researcher's best prediction of how many people would choose each alternative are 75 and 25. However, the percent correctly predicted statistic is based on the notion that the best prediction for each person is the alternative with the highest probability. This notion would predict that one alternative would be chosen by all 100 people while the other alternative would never be chosen. The procedure misses the point of probabilities, gives obviously inaccurate markets shares, and seems to imply that the researcher has perfect information.

3.8.2 Hypothesis testing

As with regressions, standard t-statistics are used to test hypotheses about individual parameters in discrete choice models, such as whether the parameter is zero. For more complex hypotheses, a likelihood ratio test can nearly always be used, as follows. Consider a null hypothesis H that can be expressed as constraints on the values of the parameters. Two of the most common hypotheses are (1) several parameters are zero, and (2) two or more parameters are equal to each other. The constrained maximum likelihood estimate of the parameters (labeled $\hat{\beta}^H$) is that value of β that gives the highest value of LL without violating the constraints of the null hypothesis H . Define the ratio of likelihoods, $R = L(\hat{\beta}^H)/L(\hat{\beta})$ where $\hat{\beta}^H$ is the (constrained) maximum value of the likelihood function (not logged) under the null hypothesis H and $\hat{\beta}$ is the unconstrained maximum of the likelihood function. As in likelihood ratio tests for models other than those of discrete choice, the test statistic defined as $-2\log R$ is distributed chi-squared with degrees of freedom equal to the number of restrictions implied by the null hypothesis. Therefore, the test statistic is $-2(LL(\hat{\beta}^H) - LL(\hat{\beta}))$. Since the log likelihood is always negative, this is simply two times the (magnitude of the) difference between the constrained and

unconstrained maximums of the log likelihood function. If this value exceeds the critical value of chi-squared with the appropriate degrees of freedom, then the null hypothesis is rejected.

Examples

Null Hypothesis I: The Coefficients of Several Explanatory Variables are Zero

To test this hypothesis, estimate the model twice: once with these explanatory variables included and a second time without them (since excluding the variables forces their coefficients to be zero). Observe the maximum value of the log likelihood function for each estimation; two times the difference in these maximum values is the value of the test statistic. Compare the test statistic with the critical value of chi-squared with degrees of freedom equal to the number of explanatory variables excluded from the second estimation.

Null Hypothesis II: The Coefficients of the First Two Variables are the Same

To test this hypothesis, estimate the model twice: once with each of the explanatory variables entered separately including the first two; then with the first two variables replaced by one variable that is the sum of the two variables (since summing the variables forces their coefficients to be equal). Observe the maximum value of the log likelihood function for each of the estimations. Multiply the difference in these maximum values by two and compare this figure with the critical value of chi-squared with one degree of freedom.

3.9 Case Study: Forecasting for a new transit system

One of the earliest applications of logit models, and a prominent test of their capabilities, arose in the mid 1970's in the San Francisco Bay Area. A new rail system, called the Bay Area Rapid Transit (BART), had been built. Daniel McFadden obtained a grant from the National Science Foundation to apply logit models to commuters' mode choices in the Bay Area and to use the models to predict BART ridership. I was lucky enough to serve as his research assistant on this project. A

sample of commuters was taken before BART was open for service. Mode choice models were estimated on this sample. These estimates provided important information on the factors that enter commuters' decisions, including their value of time savings. The models were then used to forecast the choices that the sampled commuters would make once BART became available. After BART had opened, the commuters were re-contacted and their mode choices were observed. The predicted share taking BART was compared with the observed share. The models predicted quite well, far more accurately than the procedures used by the BART consultants, who had not used discrete choice models.

The project team collected data on 771 commuters before BART was opened. Four modes were considered to be available for the trip to work: (1) driving a car by oneself, (2) taking the bus and walking to the bus stop, (3) taking the bus and driving to the bus stop, and (4) carpool. The time and cost of travel on each mode was determined for each commuter, based on the location of the person's home and work. Travel time was differentiated as walk time (for the bus/walk mode), wait time (for both bus modes), and on-vehicle time (for all the modes.) Characteristics of the commuter were also collected, including income, household size, number of cars and drivers in the household, and whether the commuter was the head of the household. A logit model with linear-in-parameters utility was estimated on these data.

The estimated model is shown in Table 3.1, which is reproduced from Train (1978). The cost of travel was divided by the commuter's wage to reflect the expectation that workers with lower wages are more concerned about cost than higher-paid workers. On-vehicle time enters separately for car and bus travel to indicate that commuters might find time spent on the bus to be more, or less, bothersome than time spent driving in a car. Bus travel often involves transfers, and these transfers can be onerous for travelers. The model therefore includes the number of transfers and the expected wait time at the transfers. The headway (i.e., the time between scheduled buses) for the first bus line that the commuter would take is included as a measure of the maximum amount of time that the person would need to wait for this bus.

The estimated coefficients of cost and the various time components provide information on the value of time. By definition, the value of time is the extra cost that a person would be willing to incur to save time. Utility takes the form: $U_{nj} = \alpha c_{nj}/w_n + \beta t_{nj} + \dots$ where c is

Table 3.1: Logit Model of Work Trip Mode Choice

Modes: 1. Auto alone, 2. Bus with walk access, 3. Bus with auto access, 4. Carpool		
Explanatory variable (Variable enters modes in parentheses and is zero in other modes.)	Coefficient	t-statistic
Cost divided by post-tax wage, in cents per minute (1-4)	-.0284	4.31
Auto on-vehicle time, in minutes (1,3,4)	-.0644	5.65
Transit on-vehicle time, in minutes (2,3)	-.0259	2.94
Walk time, in minutes (2,3)	-.0689	5.28
Transfer wait time, in minutes (2,3)	-.0538	2.30
Number of transfers (2,3)	-.1050	0.78
Headway of first bus, in minutes (2,3)	-.0318	3.18
Family income with ceiling of 7500 (1)	.00000454	0.05
Family income minus 7500 with floor of 0 and ceiling of 3000 (1)	-.0000572	0.43
Family income minus 10,500 with floor of 0 and ceiling of 5000 (1)	-.0000543	0.91
Number of drivers in household (1)	1.02	4.81
Number of drivers in household (3)	.990	3.29
Number of drivers in household (4)	.872	4.25
Dummy if worker is head of household (1)	.627	3.37
Employment density at work location (1)	-.0016	2.27
Home location in or near central business district (1)	-.502	4.18
Autos per driver with a ceiling of one (1)	5.00	9.65
Autos per driver with a ceiling of one (3)	2.33	2.74
Autos per driver with a ceiling of one (4)	2.38	5.28
Auto alone dummy (1)	-5.26	5.93
Bus with auto access dummy (1)	-5.49	5.33
Carpool dummy (1)	-3.84	6.36
Likelihood ratio index		0.4426
Log likelihood at convergence		-595.8
Number of observations		771
Value of time saved as a percent of wage:		t-statistic
Auto on-vehicle time	227	3.20
Transit on-vehicle time	91	2.43
Walk time	243	3.10
Transfer wait time	190	2.01

cost and t is time. The total derivative with respect to changes in time and cost is $dU_{nj} = \alpha/w_n dc_{nj} + \beta dt_{nj}$, which we set to zero and solve for dc/dt to find the change in cost that keeps utility unchanged for a change in time: $dc/dt = -(\beta/\alpha)w_n$. The value of time is therefore β/α proportion of the person's wage. The estimated values of time are reported at the bottom of Table 3.1. Time saved from riding on the bus is valued at 91 percent of wage $((-.0259/-.0284) \cdot 100)$, while time saved from driving in a car is worth more than twice as much: 227 percent of wage. This difference suggests that commuters consider the hassles of driving to be considerably more onerous than riding the bus, when evaluated on a per-minute basis. Commuters apparently choose cars not because they like driving *per se* but because driving is usually quicker. Walking is considered more bothersome than waiting for a bus (243 percent of wage versus 190 percent), and waiting for a bus is more bothersome than riding the bus.

Income enters the representative utility of the auto-alone alternative. It enters in a piece-wise linear fashion to allow for the possibility that additional income has a different impact depending on the overall level of income. None of the income variables enters significantly. Apparently dividing travel cost by wage picks up whatever effect income might have on the mode choice of a commuter. That is: higher wages induce the commuter to be less concerned about travel costs but does not induce a predilection for driving beyond the impact through cost. The number of people and the number of vehicles per driver in the household have a significant impact on mode choice, as expected. Alternative-specific constants are included, with the constant for the bus/walk alternative normalized to zero.

The model in Table 3.1 was used to predict the mode choices of the commuters after BART was open for service. The choice set was considered to be the four modes listed above plus two BART modes, differentiated by whether the person takes the bus or drives to the BART station. Table 3.2 presents the forecasted and actual shares for each mode. BART demand was forecast to be 6.3 percent compared with an actual share of 6.2 percent. This close correspondence is remarkable.

The figures in Table 3.2 tend to mask several complications that arose in the forecasting. For example, walking to the BART station was originally included as a separate mode. The model forecasted this option very poorly, over-predicting the number of people who would

Table 3.2: Predictions for after BART opened

	Actual share	Predicted share
Auto alone	59.90	55.84
Bus with walk access	10.78	12.51
Bus with auto access	1.426	2.411
BART with bus access	0.951	1.053
BART with auto access	5.230	5.286
Carpool	21.71	22.89

walk to BART by a factor of twelve. The problem was investigated and found to be primarily due to differences in the experience of walking to BART stations compared to walking to the bus, given the neighborhoods in which the BART stations are located. These issues are discussed at greater length by McFadden *et al.* (1977).

3.10 Derivation of Logit Probabilities

It was stated without proof in section (3.1) that if the unobserved component of utility is distributed iid extreme value for each alternative, then the choice probabilities take the form of equation (3.6). We now derive this result. From (3.5) we have

$$P_{ni} = \int_{s=-\infty}^{\infty} \left(\prod_{j \neq i} e^{-e^{-(s+V_{ni}-V_{nj})}} \right) e^{-s} e^{-e^{-s}} ds$$

where s is ε_{ni} . Our task is to evaluate this integral. Noting that $V_{ni} - V_{ni} = 0$ and then collecting terms in the exponent of e , we have:

$$\begin{aligned} P_{ni} &= \int_{s=-\infty}^{\infty} \left(\prod_j e^{-e^{-(s+V_{ni}-V_{nj})}} \right) e^{-s} ds \\ &= \int_{s=-\infty}^{\infty} \exp \left(- \sum_j e^{-(s+V_{ni}-V_{nj})} \right) e^{-s} ds \\ &= \int_{s=-\infty}^{\infty} \exp \left(-e^{-s} \sum_j e^{-(V_{ni}-V_{nj})} \right) e^{-s} ds \end{aligned}$$

Define $t = \exp(-s)$ such that $-\exp(-s)ds = dt$. Note that as s approaches infinity, t approaches zero, and as s approaches negative in-

finitly, t becomes infinitely large. Using this new term,

$$\begin{aligned}
 P_{ni} &= \int_{\infty}^0 \exp \left(-t \sum_j e^{-(V_{ni}-V_{nj})} \right) (-dt) \\
 &= \int_0^{\infty} \exp \left(-t \sum_j e^{-(V_{ni}-V_{nj})} \right) dt \\
 &= \frac{\exp \left(-t \sum_j e^{-(V_{ni}-V_{nj})} \right)}{-\sum_j e^{-(V_{ni}-V_{nj})}} \bigg|_0^{\infty} \\
 &= \frac{1}{\sum_j e^{-(V_{ni}-V_{nj})}} = \frac{e^{V_{ni}}}{\sum_j e^{V_{nj}}},
 \end{aligned}$$

as required.